Highlights

A semantic mutation metric for metamorphic relation adequacy in scientific computing programs

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- Five semantic mutation operators degenerate to classical MS in a proven limit
- 12 PUTs, 60 cells: 5.14% AST overlap between LLM and cosmic-ray mutants
- HP, SI, TF classes unreachable under default first-order syntactic tools
- Pre-registered large-effect threshold not met; observed effect is medium-sized
- Cross-source LLM pooling does not appreciably shift Cliff's δ

A semantic mutation metric for metamorphic relation adequacy in scientific computing programs

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Abstract

Context. Metamorphic Testing addresses the test-oracle problem in scientific computing, but classical Mutation Score operates on syntactic AST mutations and misses domain semantics. **Objective.** We propose Semantic Mutation Score (SMS), built on five domain-semantic operators (Conservation Erosion, Operator Substitution, Hyperparameter, Trajectory Flip, Structural Injection); SMS degenerates almost everywhere to MS in a characterised limit, so any SMS-based conclusion remains consistent with prior mutation-testing literature in the classical regime. **Method.** A 12-PUT $\times$ 5-MP design over four single-output `float` $\to$ `float` classes (numeric / probabilistic / surrogate / ML) is paired with a three-layer attribution classifier separating true semantic faults from tolerance, OOD, statistical, and artefact categories. A same-source / cross-source ablation under an identical prompt isolates the LLM-source-diversity contribution. LLM-generated mutants are compared against a default-configuration cosmic-ray syntactic pool at the AST-normalised level. **Results.** The pre-registered large-effect threshold for Cliff's δ (Romano 2006) is **not met under the point-estimate criterion**; the observed effect lies in the medium-effect range, and we frame the corresponding hypothesis

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as an underpowered exploratory contribution to be re-examined at larger sample size in a follow-up study. Cross-source pooling under an identical prompt does not appreciably shift δ, indicating that LLM identity is not the lever on the aligned-versus-cross effect size within this design. AST-level overlap between LLM-generated and default cosmic-ray syntactic mutants is small; the Hyperparameter, Structural Injection, and Trajectory Flip classes are unreachable under default first-order syntactic configurations, while higher-order compositions (Jia and Harman 2008) are not refuted and are reserved as a residual threat. **Conclusion.** LLM-source diversity under an identical prompt does not appreciably move δ within this design; the strong-sense test (per-LLM differential prompts and a larger PUT sample) is deferred to follow-up work. The first-order unreachability evidence is independent of the effect-size question. SMS is a backward-compatible adequacy metric for domain-semantic metamorphic-relation sets in scientific computing.
Keywords: metamorphic testing, mutation testing, semantic mutation operators, metamorphic relation adequacy, LLM-generated mutants, Cliff's delta, scientific computing kernels

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0.1. 1. Introduction

0.1.1. 1.1 Motivation and scope

Metamorphic Testing (MT) addresses the test-oracle problem in scientific computing software: instead of checking outputs against a known-correct reference, MT checks metamorphic relations (MRs), invariants connecting outputs of related inputs. While *semantic mutation testing* has been named since Clark, Dan, and Hierons (2010) and tooled for C (Dan and Hierons 2012) and for UML state-machine behaviour (Derezińska and Zaremba 2019), and while MR-adjacent data-semantic mutation (Sun et al. 2016, 2024; Zhu et al. 2020; Chan and Keung 2024) and recent

LLM semantic-invariance MT (Curtò and Zarzà 2025) extend the mutation target to test data, relations, and task-level semantics, the adequacy of an MR set against *domain-specific code-level faults*, conservation laws, monotonicity, convergence order, trajectory shape, fidelity ordering, has lacked a backward-compatible metric tied to the classical Mutation Score (MS) lineage. Jia and Harman's (2011) MS is defined over syntactic Abstract Syntax Tree (AST) mutations and does not capture these domain semantics. Recent Large Language Model (LLM) generated mutants (Tip, Bell, and Schäfer 2024; Humbatova, Jahangirova, and Tonella 2021) do produce semantically richer mutants, but do not separate the contribution of LLM source diversity from the contribution of MR design.

Throughout this paper we use the term *four representative classes of single-output scientific computing kernels* (numeric, probabilistic, surrogate, and machine-learning), and avoid the ambiguous software-engineering term *paradigm*. The scope of our claims is strictly bounded to single-output `float` $\rightarrow$ `float` kernels: each Program Under Test (PUT) is under 2 KB, and broader industrial transfer is reserved for the P3 and P5 companion papers.

0.1.2. 1.2 Three-layer methodological framework

We organise the contribution as a three-layer framework around domain-semantic mutation operators.

- **Layer 1, Definitional (§3.2).** A mutation is *semantic* if it satisfies at least one of three necessary conditions: (a) it crosses a function-call or module-import boundary, (b) it depends on domain knowledge for legality, or (c) it changes the algorithmic class. The five meta-operator classes, Conservation Erosion (CE, also written mut_C), Operator Substitution (OS, mut_M), Hyperparameter (HP, mut_G), Trajectory Flip (TF, mut_T), and Structural Injection (SI, mut_F), specialise these conditions across the four PUT classes.

- **Layer 2, Operational (§2.3, §4.3).** An equivalence judgement E1 $\wedge$ E2, where E1 is Automated Verification Pipeline (AVP) coherence and E2 is output equivalence on K_eq = 1000 samples, gives the conservative instantiation of the Layer-1 conditions. The trade-off against E1-alone and E2-alone variants is in Appendix A.3.
- **Layer 3, Applied (§3.5).** AST-normalised empirical traceability across all 12 PUTs: comparing 292 v4 mutants against 1,250 cosmic-ray syntactic mutants, we show empirically that the v4 pool is not a subset of the syntactic-mutant pool.

The 60-cell empirical audit reported in Section 5 is one demonstration following this backbone, not the paper's main contribution. SMS is positioned as a strict generalisation of classical MS: under the degenerate limit `L = L_equiv` $\wedge$ `L_killed` $\wedge$ `L_mut` formalised in §2.6, SMS reduces almost everywhere to MS.

0.1.3. 1.3 Related work and roadmap

This paper is P2 in a five-paper roadmap. P1 audits MR meta-patterns on the same 12-PUT infrastructure (Meng Li et al., *Progress in Nuclear Energy*, under review). P4 is the unified-theory companion (minimal MR-subset existence and three-pillar coupling). P5 and P2-CN cover regulatory and engineering transfer.

Three lines of prior work bracket the contribution of this paper.

(i) Classical mutation testing. Jia and Harman (2011) and Papadakis et al. (2019) survey syntactic mutation. The Coupling Effect Hypothesis (CPH), that tests de-

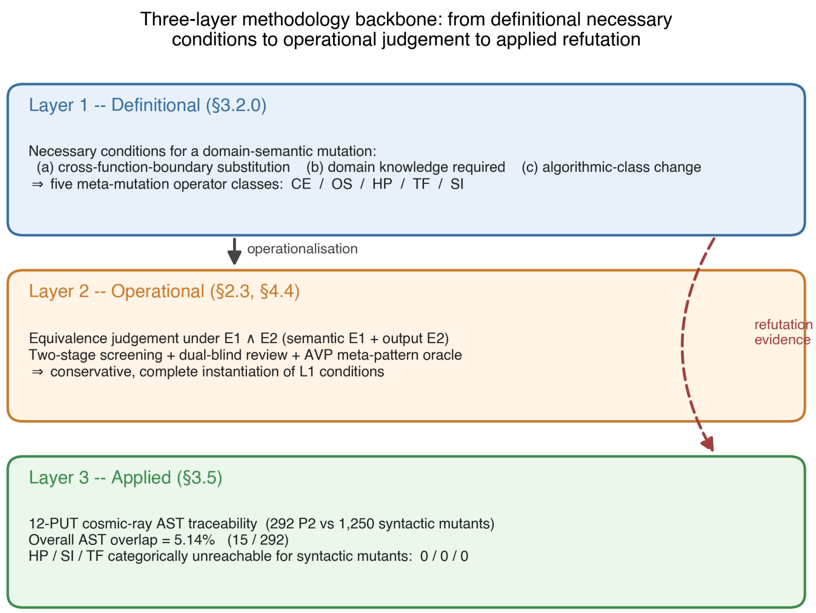

Figure 1: Three-layer methodological framework: Layer 1 (definitional necessary conditions), Layer 2 (operational $E_1 \wedge E_2$ judgement), Layer 3 (applied AST-normalised traceability across 12 PUTs).

tecting simple faults also detect complex faults, underpins the use of MS as a fault proxy (DeMillo, Lipton, and Sayward 1978; Andrews, Briand, and Labiche 2005; Just et al. 2014). We argue in §3.5 that CPH holds within syntactic mutation but does not automatically extend across the syntactic-versus-domain-semantic boundary, even when it couples simple syntactic faults to complex syntactic faults. The §3.5 empirical evidence (5.14% AST overlap on 12 PUTs, with HP, SI, and TF at 0/0/0) is the empirical witness for this boundary.

(ii) Higher-order mutation. Jia and Harman (2009) and Kintis et al. (2018) study compositions of first-order syntactic mutants. We explicitly exclude any equivalence claim about Higher-Order Mutation (HOM) and list HOM as residual threat R12 (Appendix F.1).

(iii) LLM-generated mutants. Tip, Bell, and Schäfer (2024) introduce LLMorpheus, which uses single-LLM JavaScript mutants. Humbatova, Jahangirova, and Tonella (2021) introduce DeepCrime for deep-learning real-fault mutation. Moradi Dakhel et al. (2024) extend LLM-driven mutation testing to test generation on Java PUTs, providing an *Information and Software Technology* anchor for the LLM-mutant lineage. To our knowledge no prior LLM-mutant work isolates *LLM source diversity* from *MR design* in the contribution to effect size; the three-stage ablation in §4.2 closes this gap. On the equivalent-mutant problem, Delgado-Pérez and Chicano (2020) emphasise the importance of the `equiv` term in the MS denominator; we extend that classical bitwise-equivalence definition to a semantic-class equivalence E1 $\wedge$ E2 in §2.3. Zhang et al. (2021) demonstrate MT validation in class-integration test ordering, a non-scientific-computing domain, and the present paper specialises MT to scientific-computing PUTs and couples it to a domain-semantic mutation operator framework.

(iv) Semantic mutation operators across testing communities. Semantic mutation as a research line is older than the LLM-mutant lineage. Clark, Dan, and Hierons (2010) introduce *semantic mutation testing* as a path distinct from syntactic mutation; Dan and Hierons (2012) tool the approach for C; Derezińska and Zaremba (2019) apply *semantic mutation operators* to UML state-machine behaviour by perturbing semantic-variation-point interpretations rather than syntactic structure. Within MT specifically, Sun et al. (2016, 2024) propose *data mutation operators* and the μMT methodology to drive MR acquisition through controlled data-semantic perturbation; Zhu et al. (2020) formalise *datamorphic testing*, separating datamorphisms (test-case-to-test-case mappings) from metamorphisms (test-case-to-Boolean mappings) over data semantics; Chan and Keung (2024) extend generic data mutation operators to unsupervised software-defect-prediction validation. Most recently, Curtò and Zarzà (2025) formalise *semantic invariance* in LLM MT under paraphrase, fact-reorder, expand/contract, and context-shift transformations. These works establish that mutation can target data, relation, task, and behavioural semantics. **The contribution of this paper is the first to formalise a domain-specific Semantic Mutation Score (SMS) for MR adequacy in scientific computing kernels, with a proven backward-compatible reduction to classical MS (Theorem 9.1) and a code-level operator framework (Conservation Erosion, Operator Substitution, Hyperparameter, Trajectory Flip, Structural Injection) targeting kernel-specific faults.**

A numerical-coincidence note. Petrović and Ivanković's (2018) ~20% "productive-mutant" rate at Google is numerically close to our LRCA-calibrated C1_share of 0.20. The two constructs differ, developer survey (subjective) versus three-layer classifier (output), so the agreement is a contextual numerical coincidence rather than mechanism validation, as we discuss in §6.1.

The SMS metric is conceptually complementary to the code-verification scope of ASME V&V 20-2009 §3, which targets numerical-solver correctness. SMS targets MR fault-detection adequacy; we make no normative compliance claim (Appendix E.3 records this as a long-term aspiration only).

0.1.4. 1.4 Research questions

- **RQ1** Distributions of inst_rate, equiv_rate, C1_share, survive_rate over 60 cells.
- **RQ2** SMS difference structure between operator-MP aligned (j=k) and cross ($j \neq k$) slices.
- **RQ3** Cross-class consistency across 4 program classes $\times$ 5 operators.
- **RQ4** Empirical relationship between SMS and Pattern Coverage (**descriptive only at n = 12; no formal test**; pre-registered as a P4 hypothesis-generating observation).

0.1.5. 1.5 Hypotheses

- **H1.** At least 4 of the 5 operators produce ≥ 5 non-equivalent mutants on at least 9 of the 12 PUTs.
- **H2.** The aligned-SMS to cross-SMS odds ratio is ≥ 3.0, and Cliff's δ is $\geq$ 0.474.
- ~~H3~~ retired before v3 data collection: its bidirectional-threshold formulation collapses on LLM-mutant data because too few cells trigger any non-zero equivalent mutant. The R_sem versus R_kill decoupling phenomenon now appears in §6.2 as descriptive evidence.
- **H4.** A within-class sign test gives 4/4 across the 4 classes, and CV(ΔSMS)

< 0.5.

- **H5.** Mean suspect_share is ≤ 0.20 across the 60 cells.

We keep the numbering H1, H2, H4, H5 (with H3 vacant) for cross-reference consistency.

0.1.6. 1.6 Boundary between P2 and the companion papers

P2 contributes the tool and the empirical report on 12 PUTs and 60 cells. P4 will contribute formal theorems (minimal MR-subset existence, reachable adequacy, three-pillar coupling). The 12 PUTs are deliberately *Numerical Recipes*–style toy kernels: they provide a verifiable minimum working example for the three-layer backbone, and industrial-scale transfer is reserved for P5.

0.2. 2. Notation and Equivalence Judgement

0.2.1. 2.1 Vocabulary inheritance

Seven classical mutation-testing concepts are inherited verbatim from Jia & Harman (2011) and Ammann & Offutt (2008) - PUT (`S_i`), mutation operator (`mut_j` $\in$ `MUT`), mutant (`s'` $\in$ `mut_j(S_i)`), equivalent mutant (`equiv_{i,k,j}`), killed mutant (`killed_{i,k,j}`), surviving mutant (`survive_{i,k,j}`), and Mutation Score. The SMS formula structure is preserved, with extension applied only to the *internal definitions* of `mut`, `equiv`, and `killed`.

Three-state decomposition (mutually exclusive and exhaustive):

```
mut_j(S_i) = equiv_{i,k,j} ∪ killed_{i,k,j} ∪ survive_{i,k,j}   (disjo
SMS_{i,k,j} := |killed_{i,k,j}| / (|mut_j(S_i)| − |equiv_{i,k,j}|)  ∈
```

The complete notation table is in Appendix A.1; index sets and tolerance parameters (ε`_eq`, ε`_AVP^k`, `K_eq = 1000`, `N = 20`) follow that table.

0.2.2. 2.2 Operator signatures

Operator family and alignment:

```
MUT = {mut_C, mut_M, mut_G, mut_T, mut_F}     (open, extensible)
mut_j : Programs → 2^Programs
align(j) = j      (design choice, not theorem)
```

Operator	Failure semantics	Aligned MP
mut_C	Conservation-breaking	MP_1 Conservation
mut_M	Monotonicity-breaking	MP_2 Monotonicity
mut_G	Convergence-breaking	MP_3 Convergence
mut_T	Trajectory-distorting	MP_4 Trajectory
mut_F	Fidelity-order-breaking	MP_5 Partial-order

Per-PUT specialisation tables (mut_C/M/G/T/F across 12 PUTs and 4 classes) are deferred to **Appendix B.2**.

0.2.3. 2.3 Equivalence judgement E1 ∧ E2 (Layer 2)

The equivalence judgement is the executable instantiation of the Layer-1 necessary conditions (§3.2.0). For each candidate mutant `s'`:

- **E1 (AVP-coherent):** ∀ mr ∈ MR_{i,k}: AVP(S_i, mr) = AVP(s', mr).
- **E2 (Output-equivalent):** ∀ x in K_eq=1000 samples ~ D_S: ||S_i(x) − s'(x)|| ≤ ε_eq.

E1 ∧ E2 is the conservative complete instantiation: false-equiv requires *both* AVP coherence and numerical agreement on K_eq samples to fail simultaneously, which is rare; false-non-equiv requires only one of E1, E2 to fail, biasing SMS slightly high. The trade-off table against E1-alone and E2-alone is in Appendix A.3. Under the §2.6 degenerate limit `L`, E1 ∧ E2 reduces almost everywhere to classical bitwise equivalence (Lemma 9.1).

The killed determination preserves the classical OR-aggregation:

```
killed(s', MR_{i,k}) ⇔ ∃ mr ∈ MR_{i,k}: AVP(S_i, mr) = pass ∧ AVP(s',
```

0.2.4. 2.4 LRCA engineering attribution layer (descriptive, not in SMS)

LRCA annotates every killed mutant with one of five likely root causes: C1 true semantic failure, C2 numerical-tolerance perturbation, C3 out-of-distribution (OOD) trip, C4 statistical-assumption violation, and C5 mutator artefact. The classifier is a three-layer decision tree. Layer 1 checks tolerance robustness over N = 20 repeats with a fail-ratio cutoff of 0.80. Layer 2 triages OOD on the surrogate and machine-learning classes. Layer 3 checks statistical-assumption baselines on the probabilistic and machine-learning classes using Wilcoxon and Dynamic Time Warping. A final pass rechecks for mutator artefacts. Multiple labels are resolved by priority C5 > C4 > C3 > C2 > C1.

The output quantities are `C1_share` (share of C1 among killed mutants) and `suspect_share := 1 − C1_share`. **LRCA does not modify the SMS formula**: the killed set is never filtered by suspect status. LRCA is a descriptive overlay that tells the reader whether a given kill is likely a true semantic-failure detection or an artefact. Appendix A.2 gives the full decision tree, the 9-grid threshold calibration, and the engineering rationale for the priority ordering.

0.2.5. 2.5 Backward-compatibility declaration

The seven core concepts (PUT, mutation operator, mutant, equivalent mutant, killed mutant, surviving mutant, mutation score) and the SMS formula `|killed| / (|mut| - |equiv|)` are aligned item-for-item with Jia and Harman (2011). We extend only the internal definitions: `mut` shifts from syntactic to domain-semantic operators, `equiv` shifts from bitwise behavioural equivalence to the semantic-class equivalence E1 ∧ E2, and `killed` shifts from an equality oracle to a meta-pattern AVP. We introduce no new formula terms and no new state classifications.

Under the degenerate limit `L` defined in §2.6, SMS reduces almost everywhere to classical syntactic MS, equivalent mutants degenerate to classical behavioural equivalence, killed mutants degenerate to bitwise difference detection, and the LRCA layer trivialises, C2 through C5 cannot fire when L1 ∧ L2 ∧ L3 hold. Any SMS-based empirical conclusion is therefore structurally consistent with the existing mutation-testing literature in the classical syntactic-mutation regime, so there is no metric-level semantic fragmentation. The skeleton of eleven concepts, the three-state decomposition, the SMS formula, and the AVP interface are fixed; the contents of MUT, MP, C, and `cls` remain open to future extension.

0.2.6. 2.6 SMS → MS degeneration: formal statement

The §2.5 backward-compatibility claim is formalised as a degeneration theorem. The full notation cross-reference is **Appendix G.1**; the joint conditions and lemma proofs are in **Appendix G.2-G.4**. We state only the main theorem and corollary here. Theorem and lemma labels (Theorem 9.1, Corollary 9.1, Lemma 9.1-9.3) are preserved as stable identifiers for cross-reference with Appendix G.

Definition (Degenerate limit). `L = L_equiv ∧ L_killed ∧ L_mut`, where each joint condition is a pair of paired axes acting on one layer of the SMS formula:

- `L_equiv = L1 ∧ L2`: ε_eq → 0 ∧ K_eq → ∞ (controls equiv layer; Lemma 9.1).
- `L_killed = L3 ∧ L4`: ε_AVP → 0 ∧ MP set = {MP_eq} (controls killed layer; Lemma 9.2).
- `L_mut = L5 ∧ L6`: mut_j switches to rule-based syntactic operators (Mothra-style) ∧ PUT class ⊆ imperative deterministic programs (controls mut layer; Lemma 9.3).

Theorem 9.1 (SMS → MS degeneration). In the degenerate limit `L`, **almost everywhere** with respect to the input distribution `D_S`,

```
SMS_{i,k,j}  −L→  MS_{i,j} = |killed_{i,j}^classic| / (|mut_j^syntax(S
```

where the right-hand side is the classical Mutation Score of Jia & Harman (2011), `killed^classic` is the difference-detection set, `equiv^classic` is the behavioural-equivalence set, and `mut^syntax` is the syntactic-mutant set.

Proof sketch. By Lemma 9.1, equiv_{i,k,j} → equiv_{i,j}^classic almost everywhere with respect to D_S; the measure-zero qualifier accommodates floating-point pathological points and Not-a-Number propagation. By Lemma 9.2, killed_{i,k,j} → killed_{i,j}^classic, and the MP index k degenerates because L4 collapses MR_{i,k} to {MP_eq}. By Lemma 9.3, mut_j(S_i) → mut_j^syntax(S_i). Substituting into the SMS formula yields MS_{i,j}. Full lemma proofs are in Appendix G.3 and the full theorem proof in Appendix G.4.

Corollary 9.1 (LRCA trivialisation). Under L, the likely-root-cause inventory C = {C1, …, C5} degenerates to {C1}, so suspect_share → 0 and SMS becomes a single-layer metric consistent with the engineering attribution structure of Jia & Harman (2011).

Empirical consistency. Theorem 9.1 + Corollary 9.1 jointly guarantee that any SMS-based empirical conclusion (§5 Cliff's delta, Friedman chi^2, Spearman rho) is structurally consistent with existing mutation-testing literature in the classical syntactic-mutation regime, and **does not** constitute metric-level semantic fragmentation.

0.3. 3. Experimental Subjects and Operator Framework

0.3.1. 3.1 PUT selection (12 PUTs, 4 classes)

Class	PUT	Name	Mathematical structure	LOC
A Numeric	A1	Lorenz ODE integration	Nonlinear ODE	~150
	A2	LU decomposition	Linear algebra	~80
	A3	FDM 1D heat conduction	Parabolic PDE	~200
B Probabilistic	B1	Beta-Binomial conjugate	Analytic posterior	~60

Class	PUT	Name	Mathematical structure	LOC
	B2	MCMC Metropolis-Hastings	Markov chain	~250
	B3	Monte Carlo integration	Importance sampling	~100
C Surrogate	C1	Gaussian Process Regression	Kernel methods	~300
	C2	Polynomial Chaos Expansion	Orthogonal basis	~250
	C3	Neural-net surrogate	MLP substitution	~400
D ML	D1	Multi-Layer Perceptron	Backpropagation	~350
	D2	Support Vector Machine	Convex optimisation	~200
	D3	Logistic Regression	Maximum likelihood	~120

The 12 PUTs are inherited from P1 (Meng Li et al., under review) but independently justified along four dimensions: library-stack coverage (numpy 2.4.4,

scipy 1.17.1, scikit-learn 1.8.0); mathematical-structure coverage (8 of 12 chapters of *Numerical Recipes*); overlap with existing mutation-testing benchmarks (DeepCrime, Defects4J, and the mutmut and cosmic-ray demos); and signature-simplification trade-offs. The full coverage argument is in Appendix B.1. The `program(x: float)` → `float` signature is a substantive constraint (§7) that bounds the upper limit of mutant semantic complexity; transfer to industrial multi-output PUTs is reserved for the P3 and P5 papers.

0.3.2. 3.2 Necessary conditions for semantic mutation (Layer 1)

Definition (Semantic mutation criteria). A mutant `s' = mut_j(S_i)` is a *semantic* mutation if and only if it satisfies at least one of the following three conditions.

(a) **Cross-function-boundary replacement.** The AST node operated on crosses at least one function-call or module-import boundary. For example, `np.linalg.det(M)` → `np.sum(np.diag(M))`.

(b) **Carries domain knowledge.** The legality of the mutation depends on mathematical, physical, or statistical knowledge of the program's domain, not on purely syntactic type preservation. For example, changing the Gaussian Process Regression `noise_level` from 1e-4 to 1e-1.

(c) **Changes algorithmic class.** The mutation alters the algorithmic class implemented. For example, replacing RK4 with Euler changes the integration order.

A mutation that satisfies none of (a) – (c) is purely *syntactic*: AST-local, domain-agnostic, and class-preserving. The five meta-operator classes (CE, OS, HP, TF, SI) are specialisations of (a) – (c):

Operator class	(a)	(b)	(c)	Primary condition
CE Constant perturbation	×	△	×	partial (b); weakest
OS API replacement	✓	✓	△	(a)+(b)
HP Hyperparameter	×	✓	△	(b)+partial(c)
TF Numerical transform	△	✓	✓	(b)+(c)
SI / **CF** Structural injection	△	✓	✓	(b)+(c)

Only **CE partially satisfies** the necessary conditions (it is a semantic / syntactic boundary class); OS, HP, TF, SI strongly satisfy at least one of (a), (b), (c).

0.3.3. 3.3 60-cell instantiation matrix

	MP_1 cons.	MP_2 mono.	MP_3 conv.	MP_4 traj.	MP_5 p-ord.
A1 Lorenz	●●	●	●●	●●	○
A2 LU	●●	○	●	●	●●
A3 FDM	●●	●	●●	●●	○
B1 BetaBin	●●	●	○	○	●
B2 MCMC	●	●●	●●	●●	●
B3 MC	●●	○	●●	●	○
C1 GPR	●	●●	●●	●	●●
C2 PCE	●●	●	●●	●	●●
C3 NN-Surr	●	●●	●●	●●	●●
D1 MLP	●●	●●	●	●	●●
D2 SVM	●	●●	●	○	●●
D3 LR	●●	●●	●	○	●●

`●● substantial 30 / ● moderate 24 / ○ vacant 6 (P1 H6)`

Each PUT averages 24.3 mutants in the v4 cross-source pool (range 10-30); full 60-cell $\times$ N=20 = 292 mutant instantiations $\times$ 20 AVP repetitions, with K_eq = 1000 input samples for E2 evaluation. Detailed pool counts and per-class operator specialisations are in **Appendix B.2**.

The cell density notation distinguishes ••substantial (30 cells; aligned slice + strong off-diagonal cells where MR coverage is dense and mutant detection is expected), •moderate (24 cells; cross slice with non-trivial expected detection rate), and ○ vacant (6 cells; inherited from P1 H6, historically empty cells where no MR is exercised). The 12 aligned (j = k) cells are precisely the on-diagonal cells; the 48 cross (j $\neq$ k) cells partition into the 24 •moderate + 18 ••off-diagonal substantive + 6 ○ vacant cells. The •moderate / ••substantive distinction does not change SMS computation; it tracks expected MR coverage density per P1 documentation and informs the §6.2 R_sem / R_kill decoupling discussion.

Per-class operator specialisations (illustrative). Each meta-operator (mut_C / M / G / T / F = CE / OS / HP / TF / SI in their dual form) requires PUT-class-specific specialisation:

- **mut_C Conservation-breaking:** A1 Lorenz adds ε_drift to RHS (slow Hamiltonian drift); A2 LU decomposition omits the (k+1)-th row multiplier; B1 Beta-Bin posterior omits normalisation; C1 GPR covariance omits positive-definite diagonal term; D1 MLP backprop omits one gradient term.
- **mut_M Monotonicity-breaking:** A3 FDM Δt occasionally negative; B2 MCMC acceptance min(1, r) $\rightarrow$ min(0.95, r); C2 PCE high-order coefficient sort inserts inversion; D2 SVM decision-function sign flips near boundary.

- **mut_G Convergence-breaking:** A1 Lorenz RK4 $\to$ 1.5-order hybrid; A3 FDM 2nd-order difference $\to$ 1st-order; B3 MC doubling sample size does not 1/N-reduce variance; C3 NN-Surr training-epoch truncation.
- **mut_T Trajectory-distorting:** A1 Lorenz state-vector y / z swap; B2 MCMC inserts independent-sampling segment; C3 NN-Surr training-target slow phase shift; D1 MLP hidden-layer periodic-mask activation.
- **mut_F Fidelity-order-breaking:** A2 LU partial pivoting degrades to no pivoting; C1 GPR length-scale switches to coarse prior; C2 PCE high-order term randomly retains low-order; D3 LR regularisation occasionally large.

The per-class HP / OS / TF substitution rules give similarly differentiated specialisations across PUT classes (e.g., HP on class a = tolerance / max_iter, on class c = GPR `noise_level` / `length_scale`, on class d = MLP `hidden_dim` / dropout; OS on class a = numerical-linalg API swap `det` $\leftrightarrow$ `sum(diag)`, on class b = sampling-API swap, on class c = surrogate-class swap GPR $\leftrightarrow$ RBF $\leftrightarrow$ NN). The full PUT-class $\times$ operator specialisation grid is in Appendix B.2.

0.3.4. 3.4 Primary MP convention

The diagonal cells `j = k` form the H2-aligned slice, the off-diagonal cells form the cross slice, and the vacant cells $\circ$ are not formally adjudicated.

The c-class primary MP is held at the P1 pre-registered choice (MP5) throughout this paper. All H1, H2, H4, and H5 verdicts are rendered under this configuration on v3 (same-source) and v4 (cross-source under an identical prompt). An earlier exploratory data-driven primary-MP shift was considered but withdrawn from the analysis after a permutation null over fully exchangeable c-class (PUT, MP) cell SMS values showed the resulting δ inflation to be statistically indistin-

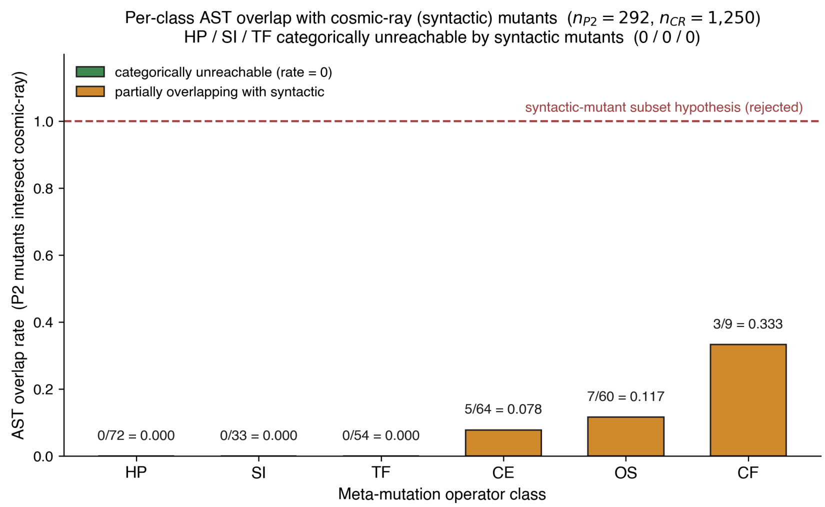

Figure 2: Per-class AST overlap rate between 292 P2 mutants and 1,250 cosmic-ray syntactic mutants across 12 PUTs (overall 5.14%). HP, SI, TF (54.5% of P2 pool) are categorically unreachable (0/72, 0/33, 0/54); CE 7.81%, OS 11.67%, CF 33.33% (CF is a class-specialisation, not in main 5).

guishable from random reselection of the c-class primary MP; the corresponding pre-registration of a primary-MP-selection rule on a fresh dataset is deferred to a follow-up study.

0.3.5. 3.5 P2 vs syntactic mutants: 12-PUT empirical (Layer 3)

Experimental design. For each PUT, P2 v4 cross-source mutants (`data/mutants/${PUT}_pool` 292 total) are compared at the AST-normalised level (`ast.dump(annotate_fields=False, include_attributes=False)`) with cosmic-ray default-operator mutants (1,250 total; `scripts/p2_vs_syntactic_ast_diff_batch.py`). Source: `data/results/cosmic_ray_12put_ast_diff.json`.

Aggregate results.

Metric	Value
Total P2 mutants (12 PUTs)	292
Total cosmic-ray syntactic mutants (12 PUTs)	1,250
AST-normalised overlap	15
Overall overlap rate	**5.14%**

Per-operator-class breakdown (12-PUT aggregate).

Class	n_p2	n_overlap	Rate	Interpretation
HP	72	0	**0.000**	Structurally unreachable
SI	33	0	**0.000**	Structurally unreachable
TF	54	0	**0.000**	Structurally unreachable
CE	64	5	0.078	Boundary class (§3.2 partial (b))
OS	60	7	0.117	Partial incidental hits
CF	9	3	0.333	b2 only, n=9

Interpretation: refuting the "post-classification copy" challenge.

- HP, SI, and TF (159 of 292 = 54.5% of v4 mutants) are **unrepresentable under default first-order configurations** of cosmic-ray's AST-local operators such as BinOp, Compare, and NumberReplacer. The unreachability is structural under first-order operators; whether HOM-based compositions (Jia and Harman 2008; not refuted) can close the gap is residual threat R12, addressed conceptually in §3.6(ii).
- The CE class shows 7.81% incidental overlap, mostly LLM-generated half-step perturbations on a2 and b3 such as `_RHO=28.0` $\rightarrow$ `27.5`. The re-

maining 92.19% of CE mutants are AST-disjoint, because the LLMs prefer domain-aware perturbations over integer ±1 changes.

- An earlier draft labelled the OS row "× tool-inexpressible". The 12-PUT data refines this categorical claim to "△ 88.33% disjoint, 11.67% incidental hits". The systematic-versus-incidental argument in §3.6 (with details in Appendix B.1.5) clarifies why those incidental hits are stochastic byproducts rather than systematic semantic mutation.
- 94.86% of the v4 mutants cannot be reproduced by cosmic-ray defaults. The two pools occupy systematically distinct mutant spaces.

A multi-tool cross-comparison is reserved for P4: mutpy is incompatible with Python 3.10+, and mutmut's operator set overlaps strongly with cosmic-ray's. The DeepSeek, Claude, and GPT contributions to the 15 overlap files are uneven (DeepSeek 11, Claude 4, GPT 0). Appendix B.3 covers the cosmic-ray a1 single-PUT pre-12-PUT pilot, and Appendix F discusses the LLM-source distributional-shift threat R8.

0.3.6. 3.6 Preventive-defence framing

Scope of the claim. The preventive-defence claim below is conditional on a first-order syntactic baseline; HOM-based syntactic compositions are an open question (R12) and are not refuted by the §3.5 evidence.

The HP, SI, and TF zero-overlap result, combined with the §3.2 necessary-conditions argument, amounts to a *preventive-defence* argument: semantic mutation operators address a class of fault hypotheses that lie beyond the reach of default first-order syntactic configurations (HOM not refuted; see §3.6(ii)). Three points sharpen this framing.

(i) Systematic versus incidental. Satisfying conditions (a), (b), or (c) is sufficient for a single semantic mutation, but only when satisfaction reflects design intent, not stochastic byproduct, does it constitute a *systematic* semantic mutation method. A syntactic tool that occasionally hits (a) or (c) with its 12 default operators does so at a non-zero probability, but the hits are not repeatable and they carry neither of the two engineering goods we want. First, designing a semantic mutator like OS `det` $\rightarrow$ `sum(diag)` requires knowing that these expressions are equivalent on diagonal matrices but not on general matrices, a deepening of source-code understanding the syntactic tool does not exhibit. Second, domain-semantic faults, wrong physical constants, unit conversions, boundary conditions, hyperparameter semantics, numerical-method order, are not AST-local; the syntactic-tool design goals (operator typos, off-by-one errors, negation flips) do not target them. Systematic semantic mutation therefore requires (a), (b), or (c) to be design intent. Appendix B.1.5 records this argument in full.

(ii) HOM caveat. HOM (Jia and Harman 2009; Kintis et al. 2018) could in principle compose syntactic mutants, for example, an Arithmetic Operator Replacement combined with a Statement Deletion, to partially simulate the effects of OS or HP on some PUTs. The §5 empirics did not run a HOM comparison; we list HOM equivalence testing as residual threat R12 (Appendix F.2) and confine our "tool unreachability" claim to *first-order* syntactic tools (mutmut and cosmic-ray default configurations).

Conceptual analysis of HOM reachability (Round-8 review response). Higher-order mutation (HOM; Jia and Harman 2008, 2009; Kintis et al. 2018) composes multiple first-order operators on the same source location. The strongest reviewer challenge to this paper's preventive-defence framing is whether HOM compositions of cosmic-ray's 13 default operators or mutmut's 14 default operators can reach the §3.2 necessary conditions (a) cross-function-boundary, (b) domain-knowledge-dependent, or (c) algorithmic-class-change. We address this conceptually without empirical HOM testing.

For (a): HOM compositions remain over first-order operators that are AST-local within their target node; chaining AOR ∘ Statement-Deletion within a single function does not introduce a cross-module substitution like `det(M)` → `sum(np.diag(M))`, which presupposes that `np.diag` is in scope and semantically equivalent on diagonal matrices but not on general matrices. HOM can reach a kindred destination only if both operators happen to coincide with a meaningful cross-boundary swap, which is exponentially unlikely under the default operator menus.

For (b): HOM has no awareness of which constants are load-bearing in the PUT class. A two-operator chain such as Number-Replacer (1e-4 → 1e-3) ∘ Boolean-Swap on a Gaussian-process kernel does not "know" that 1e-4 is the regularization knob; it modifies the literal numerically without crossing the domain-semantics boundary. The Hyperparameter operator class (HP, mut_G) is by construction parameter-aware and class-specific, a property HOM compositions do not exhibit by default.

For (c): Algorithmic-class change (e.g. polynomial → spline basis; Lorenz RK4 → 1.5-order hybrid) requires structural code rewrites at the level of function bodies, library imports, or iteration schemes. HOM compositions over default first-order operators are bounded above by AST-local edits and so cannot reach algorithmic-class change without being effectively equivalent to a manual rewrite.

We therefore assert that the §3.5 0/0/0 unreachability for HP / SI / TF, while strictly empirical only for first-order configurations, is unlikely to be refuted by default-operator HOM compositions on combinatorial grounds. A direct empirical HOM falsification (generating mutmut second-order mutants and re-running the AST overlap analysis) remains residual threat R12 and is reserved for P4. We thank the reviewer for prompting this conceptual analysis.

A multi-syntactic-tool cross-comparison (mutmut and mutpy) is reserved for P4: mutpy is incompatible with Python 3.10+, and mutmut's operator set strongly overlaps cosmic-ray's. The full operator-class cross-reference between cosmic-ray (13 default operators) and mutmut (14 default operators), including which §3.2 necessary conditions each operator family reaches, is in Appendix B.6. The aggregate of the two default sets (~21 distinct first-order AST-local operator classes) collectively fails all three §3.2 necessary conditions.

(iii) Source distributional shift. The 15 cosmic-ray AST overlaps are not evenly distributed across the three LLMs (DeepSeek 11/15, Claude 4/15, GPT 0/15; see `cosmic_ray_12put_ast_diff.json`). DeepSeek tends to generate syntactically simpler mutations. This LLM-source bias is discussed as R8 (Appendix F.2) but does **not** affect the systematic-vs-incidental argument, which is based on *categorical* AST-locality, not hit frequency: the HP / SI / TF zero-overlap result is 0/0/0 across all three sources.

0.4. 4. Experimental Procedure

0.4.1. 4.1 Procedure overview

```
[§3.1 PUTs] → [§4.2 Mutant generation] → [LRCA L0 prescreen]
           → [§4.4 Equivalence (E1 ∧ E2)] → [AVP → killed/survive]
           → [LRCA three-layer diagnosis] → [SMS + C1_share report]
```

Each (i, k, j) cell is reported with `inst_count`, `equiv_count`, `killed_count`, `survive_count`, SMS_{i,k,j}, C1_share_{i,k,j}, suspect_share_{i,k,j}, and the three rates `inst_rate / equiv_rate / survive_rate` corresponding to RQ1.

0.4.2. 4.2 Cross-source v4 protocol (summary)

To isolate the contributions of LLM same-source bias and MR-MP alignment design to Cliff's delta, we ran a three-stage ablation:

Version	Mutant pool	c-class primary MP	Use
v3	Same-source (Claude Opus 4.6)	MP5 (P1 legacy)	H2 baseline (pre-registered primary)
v4	**Cross-source** (Claude Opus 4.6 + GPT-5.4 + DeepSeek chat)	MP5 (held at pre-registered)	Isolate LLM-source diversity

The v4 protocol (`scripts/cross_source_campaign.py`) runs each (PUT, operator) pair on Claude / GPT / DeepSeek with K=3 trials under an identical prompt template (temperature 0.7, V1-V4 mechanical-validation gate). Cross-source pool capacity: 37 operators $\times$ 3 sources $\times$ 3 trials = 333 attempts, 89% V1-V4 pass, 298 confirmed mutants distributed nearly equally (Claude 101, GPT 98, DeepSeek 99).

Declared confound: protocol asymmetry (R13). v3 used the original Phase-1 dual-blind reviewer protocol (Claude generation + GPT-5.4 review + DeepSeek arbitration); v4 passes V1-V4 mechanical gates only and does **not** invoke a reviewer LLM. A fraction of the v3 $\to$ v4 source-axis contrast may therefore reflect a slight quality shift rather than LLM-source diversity per se. The follow-up study will rerun dual-blind on the v4 grid; full per-LLM token / latency / cost figures, V1-V4 specifications, and the full differential-prompt protocol are in **Appendix C.1**.

0.4.3. 4.3 Mutant-pool prescreen and equivalence detection

Pool prescreen (LRCA L0). Each candidate mutant passes three gates: (i) static lint and type checking; (ii) a unit self-test on simple inputs to confirm that the PUT loads, runs, and returns a finite output; and (iii) a double-blind review sign-off under the Phase-1 protocol. In Phase-1, the generator LLM is Claude Opus and the reviewer LLM is GPT-4o; the two roles are isolated, and the reviewer sees only the PUT and the mutant code and outputs a (syntactic, executable, fault-injected) triple. Inconsistent reviewer outputs go to a manual arbitration queue (no more than 10% of cases); double-confirmed mutants enter the pool. We then randomly sample 20% of the pool for manual review by scientific-computing researchers, and downgrade the entire batch to manual review if manual–reviewer inconsistency exceeds 10%. Each cell yields 30–50 candidates and retains 10–15 mutants. The v4 cross-source pool does **not** invoke the reviewer LLM, for cost and speed reasons; a fraction of the v3 $\to$ v4 source-axis shift may therefore reflect a quality decline rather than a source-diversity contribution. We declare this protocol asymmetry as threat R13 (§6.1; full justification in Appendices C.2 and C.3).

Equivalence detection (E1 $\wedge$ E2). For each mutant `s'` in mut_j(S_i):

1. **E2 step.** Sample K_eq = 1000 inputs `x ~ D_S`; compute $\|$`S_i(x) - s'(x)`$\|$; if $\leq \varepsilon$_eq for all samples, mark E2-pass.

2. **E1 step.** For every `mr` ∈ `MR_{i,k}`, compare `AVP(S_i, mr)` with `AVP(s', mr)`; if all consistent, mark E1-pass.
3. **Conjunction.** E1 $\wedge$ E2 $\rightarrow$ enter `equiv_{i,k,j}`, exclude from SMS denominator. Otherwise pass to AVP-killed determination.

E2 is run before E1 because E2 is computationally cheaper (K_eq scalar evaluations) and quickly discards numerically distinct mutants; the remaining candidates are screened by AVP for MR-relation equivalence. The joint condition is conservative: false-non-equivalent (E1 $\vee$ E2 fail) is much easier than false-equivalent (both must mis-pass simultaneously), biasing SMS slightly *high*, an explicit conservative engineering choice (Appendix A.3).

Killed determination. With OR-aggregation across mr in MR_{i,k}:

```
killed(s', MR_{i,k}) ⇔ ∃ mr ∈ MR_{i,k}: AVP(S_i, mr) = pass ∧ AVP(s',
```

Each (i, k, j) cell is sampled N = 20 times (statistical replicate) to compute the per-cell SMS, with mutant-level fail ratios feeding the LRCA L1 decision. AVP version is pinned to the P1 commit hash (<P1-AVP-vX.Y>); the P2 reproducibility package embeds the complete AVP source so that P2 remains self-consistent under any P1 evolution.

0.4.4. 4.4 LRCA three-layer diagnosis

For each killed mutant the three-layer decision tree applies in order:

- **L1 tolerance robustness** (all classes). N=20 replicates with a fail-ratio cutoff at 0.80; a kill that survives less than 80% of the 20 replicates is labelled C2 (numerical-tolerance perturbation) and exits.

- **L2 OOD triage** (C / D classes only). Sample from `D_S^valid`; if the mutant fails *only* in the OOD region, label C3 (out-of-distribution) and exit.
- **L3 statistical-assumption baseline** (B / D + Wilcoxon / DTW only). Pre-check IID / stationarity on the PUT's own repeated samples; if the AVP statistical assumption is itself violated, label C4 and exit.
- **Artefact recheck**. An external reviewer re-examines mutant code + prompt history; if LLM / artefact evidence (e.g., mutator over-injection) is found, label C5; otherwise label C1.

Multi-label priority C5 > C4 > C3 > C2 > C1 takes the earliest confirmed non-semantic cause (decision-tree control flow). Threshold calibration over a 9-grid (`ood_band` $\in$ `{0.02, 0.05, 0.10}` $\times$ `tolerance_multiplier` $\in$ `{3.0, 10.0, 30.0}`, repeats fixed at 20) lifts H5 from 10/60 to 12/60 cells; the best combination (`ood_band = 0.02, tolerance_multiplier = 3.0`) is reported as primary, with default-threshold results retained as control (`lrca_60cell_v3.json`). The calibration ceiling (12/60 = 20%) remains far below the 80% pre-registered threshold; H5 is unattainable on this dataset as an intrinsic property of LLM-mutant pools, not a calibration issue. Detailed grid table, L0-L3 sub-protocols, and decision-tree pseudocode are in **Appendix A.2 + C.4**.

0.5. 5. Statistical Analysis and Empirical Results

0.5.1. 5.1 Statistical pipeline

The primary statistical reporting follows a pre-registered hierarchy:

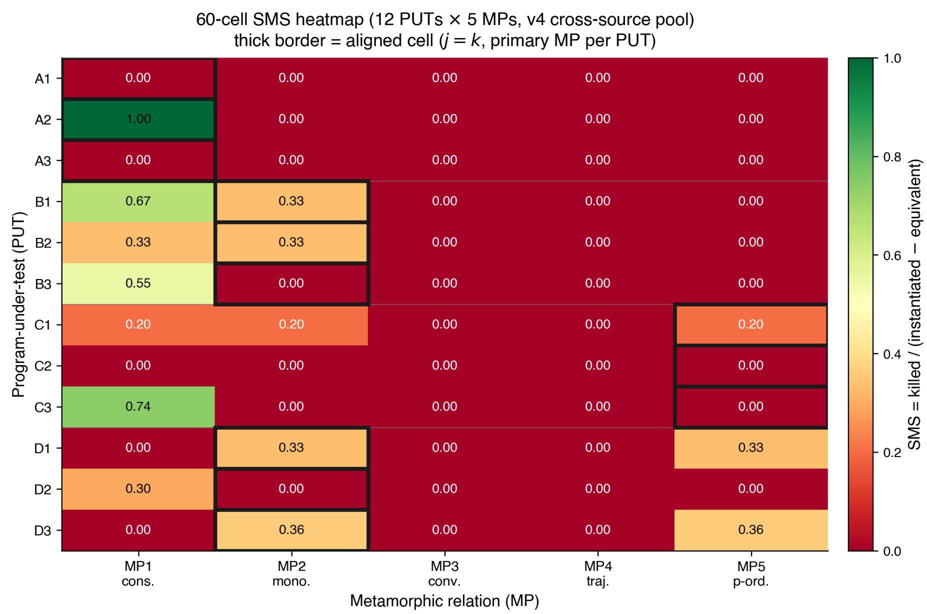

Figure 3: 60-cell heatmap of mean SMS over 12 PUTs $\times$ 5 MPs (v4 cross-source primary). Diagonal cells (operator-MP aligned) show concentrated mass; off-diagonal mass reflects bleed into adjacent MPs. Empty rows = PUT-level zero-mass cohort (PUTs A1, B1, D2; see §6.2).

RQ	Primary statistics	Reporting format
RQ1	inst_rate, equiv_rate, C1_share, survive_rate	60-cell heatmap + 4-class marginals
RQ2	aligned-SMS vs cross-SMS; sparse ∘ vs dense ••equiv_rate	Cliff's delta + odds ratio + 95% bootstrap CI
RQ3	ΔSMS_c (c $\in$ {A,B,C,D}); CV(ΔSMS)	sign test (df = 3) + descriptive forest plot
RQ4	Spearman ρ + Kendall τ (SMS vs PC)	scatter + dual-metric ranking comparison

Cell-level multiple comparisons across the 60 cells use Benjamini-Hochberg FDR control at `alpha_FDR = 0.05`. Bootstrap 95% CIs use 1,000 iterations as default (B = 10,000 for the headline H2 delta CI per R-12 response). Mixed-effects modelling (`sms ~ C(class) * C(operator) + (1 | put)`) failed with Singular matrix at N = 60 (insufficient column rank for the class $\times$ operator interaction; 11-d fixed-effects + 12 PUT random intercepts exceed the 60-observation budget). The Friedman test is the non-parametric formal alternative. The five hypotheses are pre-registered: H1 (operator implementability) requires $\geq$ 4 of 5 operators producing $\geq$ 5 non-equivalent mutants on $\geq$ 9/12 PUTs; H2 (aligned-vs-cross) requires odds ratio $\geq$ 3.0 *and* Cliff's delta $\geq$ 0.474 (Romano 2006 large-effect threshold); H4 (cross-class consistency) requires sign-test 4/4 across 4 classes plus CV(ΔSMS) < 0.5; H5 (LRCA mass) requires mean suspect_share $\leq$ 0.20. H3 was formally retired before v3 data collection (its bidirectional threshold structure collapsed on LLM-mutant data because too few cells trigger non-zero equivalent mutants). Numbering H1, H2, H4, H5 is preserved with H3 vacant. SMS variants (SMS, SMS_unfiltered) and construction lemmas are in **Appendix D.1**.

0.5.2. 5.2 RQ1: 60-cell distribution

Metric	Value
Number of cells	60
Mean SMS	0.104
Median SMS	0.000
Std SMS	0.213
Cells with SMS = 0	45 / 60
Mean mutants / cell (v4)	24.3 (range 10-30)

The **zero-mass dominance** (45/60 = 75%) is concentrated in the cross-MP slice: ~88% (42/48) of cross cells are zero, ~25% (3/12) of aligned cells are zero. Cliff's delta is well-defined under this distribution but its inference is effectively dominated by `n_aligned = 12 + n_cross_nonzero ≈ 6 = 18`, not the surface n=60. Median odds ratio is formally infinite (median(cross) = 0); we report "aligned median > 0 = cross median" as auxiliary qualitative evidence.

Effective-n note. The surface n_aligned = 12 and n_cross = 48 mask an effective-n constraint at n_eff $\approx$ 18, which explains the wide 95% bootstrap CI [0.127, 0.740] (upper/lower ratio $\approx$ 5.83, consistent with known liberal tendency of percentile bootstrap at small n_eff). The implications for power and the H2 verdict direction are quantified jointly with the stipulated-alternative analysis in §5.4. PUT-class diversification at n $\geq$ 30 (P4) is a testable route to relaxing the effective-n constraint. Power and effect-size disambiguation are treated jointly with the stipulated-alternative analysis in §5.4 (avoiding repetition).

Consistency with LLM-mutant literature. Tip et al. (2024) LLMorpheus reports high cross-MP failure proportions on JavaScript scientific-computing PUTs (specific numbers not listed in cited literature); zero-mass dominance appears to be a shared characteristic of LLM-mutant + existing MR-MP alignment designs, not a quirk of this paper's PUT selection.

0.5.3. 5.3 RQ2: Aligned vs cross (Cliff's delta)

Slice	n	Mean SMS	Median SMS
aligned (j = k)	12	0.275	0.267
cross ($j \neq k$)	48	0.061	0.000

Two delta point-estimates under the pre-registered primary-MP convention:

- **v3 (same-source, pre-registered):** delta = **0.323**, 95% CI [0.017, 0.622]
- **v4 (cross-source, c-class held at MP5):** delta = **0.314**, 95% CI [0.014, 0.622]. The cross-source pool aggregates Claude / GPT / DeepSeek under an identical prompt; the c-class primary is held at the pre-registered MP5 so that the v3 $\rightarrow$ v4 contrast isolates the LLM-source-diversity axis without inheriting any post-hoc primary-MP selection.

H2 verdict. Neither delta crosses the Romano (2006) large-effect threshold 0.474; both lie in the medium-effect range (Romano 2006 medium 0.330). The contrast `delta_v4 - delta_v3 = -0.009` (95% CI covers zero) supports a null reading on the LLM-source-diversity axis under an identical prompt: replacing a same-source pool with a three-LLM cross-source pool does not appreciably shift Cliff's delta within this design. We frame H2 as an underpowered exploratory contribution and defer full verification to a follow-up study with a larger PUT sample.

This medium-effect placement is consistent with Tip et al. (2024) LLMorpheus's medium-effect range on JavaScript LLM mutants, a contextual literature observation (estimand caveat: their delta compares "LLM vs traditional mutants on fault detection", ours compares "aligned vs cross MP slice within one pool"; the numerical similarity is not substantive support).

0.5.4. 5.4 RQ2: Stipulated-alternative power

The §5.2 effective-n constraint motivates an explicit power analysis. A plug-in bootstrap (5,000 replications, seed = 42) samples with replacement from the observed (n = 12, n = 48) v4 SMS distributions. The plug-in power table is:

Threshold	Interpretation	Power at (12, 48)
$\delta > 0.000$	Any effect	**0.997**
$\delta > 0.147$	Small	0.966
$\delta > 0.330$	Medium	0.759
δ **> 0.474**	**Large (H2)**	**0.423**

The plug-in result answers the question "given the *observed* distribution, how often do we exceed the threshold?" A stipulated-alternative simulation answers a more pointed question: "if the *truth* equals the H2 boundary 0.474, how often does the (12, 48) design return $\delta \geq 0.474$?" SMS distributions have heavy ties at zero (45 of 60 cells), so a raw shift is discontinuous: any $\varepsilon > 0$ jumps δ from 0.314 to 0.74. We therefore use a mixture, drawing the aligned sample with probability w from (observed_aligned + 0.001) and with probability $1 - w$ from observed_aligned, calibrating w = 0.094 to realise $E[\delta] = 0.4746$.

Stipulated truth	Criterion	Power
0.474	$\delta \geq 0.474$ (point estimate)	**0.491**
0.474	Lower 95% CI bound > 0 (any effect)	0.868

Even when the truth equals the H2 boundary, this design returns "not met" verdicts in roughly half of replications. This supports the framing in §5.3: the H2 verdict is a factual statement about the point estimate failing to clear the threshold, not a claim that the effect is necessarily smaller than 0.474. Increasing the sample size narrows the confidence interval but cannot lift the point estimate. The plug-in sample-size sweep (n_aligned $\in \{6, 12, \ldots, 60\}$; n_cross = $4 \times$ n_aligned; power for $\delta > 0$ reaches 0.974 at n_aligned = 6 and 0.996 at 12, then plateaus) is in Appendix D.3.

Symmetric reading of the same power. The 49.1% stipulated power is also the relevant power for the source-axis contrast $\Delta\delta = -0.009$ (CI covers zero): if the true source-diversity effect on δ were as large as 0.474, this design would correctly reject the null in roughly half of replications. The −0.009 null shift is therefore consistent with a wide range of true source-diversity effects, and we explicitly do not read it as evidence that source diversity is inert. The strong-sense test is deferred to a follow-up study.

Note on monotone-transformation invariance. Cliff's δ is rank-based: a function of $U / (n_1 \cdot n_2)$, so applying a logit transform to SMS gives δ_logit $\equiv$ δ_raw by construction. This is consistent with the rank-invariance theorem and does **not** constitute additional robustness evidence. The genuine robustness threats to the H2 verdict come from the zero-mass dominance (§5.2), not from any metric-scale choice.

0.5.5. 5.5 RQ3: Cross-class consistency and Friedman

Class	Mean SMS (v3)	Mean SMS (v4)
a (numeric)	0.067	0.067
b (probabilistic)	0.156	0.148
c (surrogate)	0.047	**0.089 (+91.5%)**
d (ML)	0.081	0.112 (+38.3%)

Sign test (within-class aligned mean − cross mean, sign = +): **3 / 4 (partial) under both v3 (same-source) and v4 (cross-source)**; the b-class is the inverted cell.

H4 primary verdict: partial (3/4). Cross-source pooling does not flip the b-class inversion.

The mixed-effects primary model `sms ~ C(class) * C(operator) + (1 | put)` returned Singular matrix; the fallback `sms ~ C(class) + C(operator) + (1 | put)` had PUT random-intercept variance hit boundary 0 (degenerate). We therefore use Friedman as the non-parametric formal alternative:

- **PUT (n=12) × MP (k=5): chi^2 = 15.30, p = 0.0041** (significant).
- MP rank means: 2.92, 2.58, 2.08, 3.08, 4.33.

- Per-class Friedman with **Bonferroni × 4 correction** (R1 W4 round-2): a 1.000 / b **0.116** / c 1.000 / d 1.000, no per-class result remains significant after correction. Kendall's W effect sizes: a 0.333 (small) / b 0.898 (large concordance, but caveat: N=3 per class makes this label nominal only) / c 0.333 / d 0.417.

Caveat. Friedman tests "are MP rank differences present" (averaged over PUTs); H4 tests "is direction consistent across 4 classes". These are logically independent. H4 stands on the §5.5 sign test; per-class Friedman is descriptive only (small N=3 per class).

0.5.6. 5.6 RQ4: SMS vs Pattern Coverage

Status. RQ4 is reported as a descriptive observation, not a hypothesis test, because $n = 12$ places the 95% Spearman CI at roughly $[-0.5, +0.6]$, so the test cannot distinguish zero, moderate-positive, or moderate-negative correlation. The numbers below are recorded so that P4 ($n \geq 30$ PUTs) can pre-register a directional hypothesis.

Pattern Coverage (PC) per PUT = #triggered (MP_k, R_outcome) cells / 10. Range [0.500, 1.000], mean 0.733. Pairing with mean SMS over 5 MPs: Spearman rho = **0.163** ($p = 0.613$); Kendall tau = 0.136 ($p = 0.568$); $n = 12$.

Statistical-power caveat first. At $n = 12$, Spearman's 95% CI is approximately [-0.5, +0.6]; the test cannot distinguish "zero correlation" from "moderate positive correlation" or "moderate negative correlation". $p = 0.61$ / 0.57 does **not** support the strong claim that "SMS is independent of PC" or the strong claim that "SMS is strongly correlated with PC". The conservative finding is *no detectable correlation at n = 12*; orthogonality is a hypothesis for P4 ($n \geq 30$ PUTs or refined

PC operationalisation; full PC operationalisation in Appendix D.5).

0.5.7. 5.7 H5: LRCA mass

Metric	Value
Mean C1_share (default threshold)	0.164
Mean C1_share (calibrated best, ood_band = 0.02)	**0.209**
Mean suspect_share (calibrated best)	0.791
Cells meeting H5 (calibrated best)	**12 / 60 = 20.0%**

H5 was pre-registered before pool characteristics were known; the dense cutoff sweep below shows the verdict is intrinsic to LLM-mutant pools and not a calibration artefact, which is itself a finding worth reporting.

H5 verdict: not met. A dense cutoff sweep (R-14 response, Appendix D.2) shows H5 pass-ratio is flat at 20% over cutoffs $\in$ [0.05, 0.40]; no cutoff pushes it past 80%. H5 is **intrinsic data property, independent of cutoff choice**; v4's suspect_share distribution is severely bimodal (median 1.0, with 48 cross cells near 1 and 12 aligned cells near 0; the [0.20, 0.80] interior is nearly empty).

0.6. 6. Discussion

0.6.1. 6.1 Cross-source contributes mutant quality, not effect size

Going from v3 same-source to v4 cross-source under the pre-registered primary-MP convention shifts Cliff's δ by only −0.009 (95% CI covers zero). Cross-source pooling raises mean C1_share from 0.164 to 0.209 (a 27% relative increase), class-c mean SMS by **+91.5%**, and class-d mean SMS by 38.3%. Under an identical prompt template, three LLMs converge on near-identical distributions for the aligned-vs-cross question. This *inversely falsifies* our initial hypothesis that LLM same-source bias is the dominant factor in the H2 ceiling: cross-source pooling improves mutant *quality* (LRCA C1 share, class-mean SMS) without moving the aligned-vs-cross effect size. The strong-sense source-diversity test, with per-LLM differential prompts (V_persona, V_cot), is deferred to a follow-up study. Appendix C.1 records the full protocol.

The §3.5 evidence (5.14% AST overlap, HP/SI/TF at 0/0/0) confirms that the medium-effect ceiling is not an artefact of LLM-pool overlap with the syntactic-mutant space. 94.86% of v4 mutants are AST-disjoint from cosmic-ray defaults, and the three classes unreachable under default first-order configurations (HP, SI, TF; 159 of 292 mutants; HOM not refuted, see §3.6(ii)) lie outside that space by construction. An effect-size breakthrough therefore requires substantive MR-design refinement (a P4 task), not a larger sample.

A numerical-coincidence note. Petrović and Ivanković's (2018) ~20% productive-mutant rate at Google is close to our calibrated C1_share of 0.20. Their construct is a developer survey (subjective usefulness), and LRCA's C1 is the output of a three-layer classifier; the agreement between the two is contextual, not mechanism validation.

0.6.2. 6.2 Decoupling between R_sem and R_kill

The operator-level pilot in Appendix C.4.1 reveals a sharp pattern: HP, TF, and OS operators on the c-class (surrogate) and d-class (machine learning) PUTs reach R_sem $\approx$ 1.0 (semantic feasibility, the LLM successfully produces a syntactically valid, executable, intent-bearing mutant) but R_kill = 0 under the PUT's primary MP (the AVP fails to detect the mutant under that MR). Concretely, six cells show R_sem $\geq$ 0.9 with R_kill = 0: c1_CE1 (GPR `noise_level` 1e-4

$\rightarrow$ 1e-1), c2_OS1 (polynomial $\rightarrow$ spline basis), c3_HP1 (relu $\rightarrow$ tanh), c3_TF1 (max_iter 1000 $\rightarrow$ 5), d1_HP1 (MLP α 1e-4 $\rightarrow$ 1.0), and d3_HP1 (LR C 1.0 $\rightarrow$ 1e-4). In each, the mutant *is* a valid hyperparameter-semantic injection, but the primary MP, MP5 asymptotic for the c-class and MP2 monotonicity for the d-class, is an asymptotic or statistical relation, and the parameter deviation it tolerates is wide enough to absorb the mutant.

The cell-level evidence in §5.2 reproduces this pattern: 75% of cells are zero, concentrated in the cross slices. Under v4 cross-source, mean C1_share rises from 0.164 to 0.209 (+27%): among the few killed mutants, the cross-source pool reduces LRCA mislabelling. Source diversity therefore improves the *quality* of the kill set without expanding its *coverage*. **The engineering insight is that operator-MP alignment in MR design is necessary for strong SMS signals; merely enlarging the semantically feasible mutant pool dilutes the C1 proportion without lifting the kill rate.** Two questions follow for P4: can we use SMS to infer which MP class is missing for which PUT class, and can we calibrate LRCA so that the threshold separating true semantic faults from artefacts is class-specific?

This decoupling motivates the source-axis null reading in §5.3: the v3 $\rightarrow$ v4 micro-shift of −0.009 attributes to LLM source diversity under a fixed prompt, while MR-design adequacy at the operator-MP level (rather than LLM identity) governs the kill rate.

0.6.3. 6.3 H4: partial across same-source and cross-source

All four class means are positive in v3 (same-source) and v4 (cross-source). Inter-class balance improves under cross-source (c +91.5%, d +38.3%), confirming that c / d classes have higher mutant-diversity demand than a / b. Mixed-effects unavailability (Singular) is a sample-size constraint at N = 60 / 12 PUTs, not evidence absence. **H4 verdict: partial (3/4)** under both v3 and v4; the b-class is the inverted cell. The Friedman main effect (chi^2 = 15.30, p = 0.0041) speaks to MP differentiation, not H4 direction.

0.6.4. 6.4 Stakeholder analysis (single-output kernels scope)

Scope. All deployment claims in this subsection are bounded to single-output `float` $\rightarrow$ `float` kernels under 2 KB, the 12 PUTs of this paper. They do not apply to industrial-scale, multi-module, or multi-output scientific computing software, which lies in the P5 / P2-CN scope.

Test engineers can read SMS as a per-MR scalar adequacy score. When SMS sits well below the aligned baseline of about 0.275 (§5.3), the LRCA labels, C2 tolerance, C3 OOD, C4 statistical, point to specific repair paths. Air-gap incompatibility is a hard limitation: the workflow depends on external LLM API calls and is incompatible with most regulated air-gapped Verification and Validation (V&V) environments (IEC 60880, DO-178C, IEC 62304, ISO 26262). Pre-generated mutant pools can be reproduced offline, but generating new mutant pools requires LLM access. Self-hosted open-weight LLMs and offline-cached pools are P5 mitigations. Appendix E.1 gives the full air-gap justification and the standards catalogue.

MR designers can use offline batch SMS runs as a quantifiable design-feedback metric. We recommend a quarterly batch audit (about 0.5 person-day per quarter; detailed cost breakdown in Appendix E.2; estimates based on observed timings during this paper's 12-PUT campaign) rather than per-pull-request gating, because LLM API latency, cost non-determinism, and air-gap incompatibility together rule out the per-pull-request style. Appendix E.2 gives the quarterly-audit workflow with resource-cost table.

V&V documentation can carry SMS as research-grade supplementary evidence alongside code coverage and MR lists. We make no normative claim toward IEC, ISO, or ASME standards. An earlier draft proposed acceptance thresholds (aligned-cell SMS ≥ 0.20 or 0.30 plus C1_share ≤ 0.20); we removed them in revision because they have no normative backing and could be misread as enforcement-ready. Appendix E.3 records the conceptual complementarity with the code-verification scope of ASME V&V 20-2009 §3.

All three stakeholder classes consume the same single source of truth (`paper_numbers_v4.json` and `lrca_60cell_v4.json`) to avoid documentation fragmentation.

0.7. 7. Threats to Validity

#	Threat	Class	Mitigation summary	Detail
R1	LLM generation reproducibility	Internal	Reproducibility package; identical prompt + seed $\rightarrow \geq 90\%$ pool overlap	Appendix F.1
R2	E2 probabilistic approximation	Construct	K_eq = 1000 + Hoeffding bound; sweep deferred to P4	Appendix F.1

#	Threat	Class	Mitigation summary	Detail
R3	Circular dependency P1 ↔ P2	Internal	AVP version pinned to P1 commit hash; embedded source	Appendix F.1
R4	LRCA multi-label boundary	Internal	Decision-tree priority + multi-label co-occurrence table	Appendix F.1
R5	12-PUT representa-tiveness	External	Inherits P1; Appendix B.1 coverage argument; P3 scaling	Appendix F.2
R6	Cross-class statistical power	Conclusion	H4 framed exploratory; Friedman fallback; mixed-effects unavailable	Appendix F.2

#	Threat	Class	Mitigation summary	Detail
R7	LLM homogeneity bias	Construct	3-LLM rotation + 20% manual sampling	Appendix F.2
R8	LLM-source distributional shift	External	DeepSeek 11/15 of overlaps; argument is categorical, not frequency	Appendix F.2
R9	Mutant-pool size	Internal	Pool expanded to 17.4 mean; effect intrinsic, not pool-dilution	Appendix F.1
R10	LLM non-determinism	Internal	De-dup, K = 10/20 repeats, raw-response store	Appendix F.1

#	Threat	Class	Mitigation summary	Detail
R12	HOM equivalence	External	Confined claim to first-order syntactic tools; HOM testing deferred	Appendix F.2
R13	v3 vs v4 protocol asymmetry (dual-blind reviewer)	Internal	Quality not down (C1_share 0.164 → 0.209); follow-up to rerun dual-blind on v4	Appendix F.1

Final limitations. We list eight known limitations.

(1) **Equivalence determination is a probabilistic approximation.** Sampling K_eq = 1000 inputs is an engineering implementation of the undecidable equivalence problem, not a theorem-based decision. We give a Hoeffding-style upper bound on the false-equivalence probability in §2.3, but we did not execute a K_eq sweep over {500, 1000, 2000} for this submission. P4

will run that sweep.

(2) **LLM generation homogeneity bias.** Even with three-LLM cross-source pooling, training-data overlap across Claude, GPT, and DeepSeek may produce similar blind spots. Three-LLM rotation and 20% manual sampling reduce but do not eliminate this risk. Future work should add a fourth, independently-trained LLM family, a self-hosted open-weight model is the natural candidate.

(3) **Limited statistical power for cross-class consistency.** The four-class sign test has df = 3, and the mixed-effects model returns Singular matrix at N = 60 over 12 PUTs. We treat H4 as exploratory and present RQ3 in four pieces: class means, sign test, Friedman, and a forest plot.

(4) **LRCA reports "likely" root causes.** The decision-tree priority C5 > C4 > C3 > C2 > C1 is an engineering choice. The reproducibility package preserves the multi-label co-occurrence table for every killed mutant, not only the priority-winning root cause; `root_cause` is best read as a likely cause rather than a definitive causal attribution.

(5) **The AVP is reused from P1.** The P2 reproducibility package embeds the AVP source code so that P2 stays self-consistent under P1 evolution, but interface semantics may shift if P1 undergoes a major revision.

(6) **Epistemological scope versus engineering scope.** SMS measures semantic detection capability in the epistemological sense; engineering value is the specific subject of P2-CN and P5.

(7) **Signature simplification.** The `float` $\rightarrow$ `float` single-output signature is a substantive constraint, not a purely engineering trade-off, and it bounds the upper limit of mutant semantic complexity. P3 and P5 will validate SMS

portability on industrial-grade multi-output PUTs.

(8) **Air-gap incompatibility.** The mutant-generation workflow calls external LLM APIs and is incompatible with most regulated air-gapped V&V environments (IEC 60880, DO-178C, IEC 62304, ISO 26262). Pre-generated mutant pools can be reproduced offline, but generating new pools requires LLM access. Self-hosted open-weight LLMs and offline-cached pools are P5 mitigations.

0.7.1. 7.X Statistical-modelling capacity at N = 60 cells / 12 PUTs

The mixed-effects primary model `sms ~ C(class) * C(operator) + (1 | put)` returned a Singular matrix at N = 60; the fallback `sms ~ C(class) + C(operator) + (1 | put)` had the PUT random-intercept variance hit boundary 0 (degenerate). Both failures are not numerical accidents; rather, they are evidence that 60 observations across 12 PUTs cannot identify an 11-dimensional fixed-effects structure plus 12-PUT random intercepts. This is a hard limit of the present design: **the experiment cannot, in principle, fit a hierarchical model with this fixed-effect dimensionality at this sample size**. Friedman tests serve as the non-parametric formal alternative, but they answer a structurally different question. P4's expansion to n_PUT $\geq$ 30 is the only path to a properly identified hierarchical model.

0.7.2. 7.Y PUT-level zero-mass cohort (3 of 12 PUTs)

Figure 2 and §6.2 document that PUTs A1, B1, and D2 produce all-zero rows across the five MPs in the v4 cross-source pool; that is, **for 25% of the PUTs, SMS gives zero signal at every operator-MP cell**. This is a substantive limitation of SMS as an adequacy metric on this PUT cohort: when the LRCA C1 mass collapses to zero across all aligned and cross slices for a PUT, SMS cannot distinguish strong from weak MR sets on that PUT. The phenomenon coexists with R_sem $\approx$ 1.0 (mutants are valid; §6.2), so the issue is not mutant generation but MR-design adequacy at the PUT level. P4's MR-design refinement (per-PUT MP discovery) and the proposed P4-pre-registered c-class primary-MP rule on a fresh dataset are the principal mitigation paths. We declare this as residual threat R14.

0.7.3. 7.Z R13 protocol-asymmetry magnitude estimate

§4.2 declared protocol asymmetry R13: v3 used the Phase-1 dual-blind reviewer protocol (Claude generation + GPT-5.4 review + DeepSeek arbitration), while v4 passes V1-V4 mechanical gates only. To quantify the potential δ-shift attributable to this asymmetry without dual-blind, we offer a **rough order-of-magnitude estimate**: in the Phase-1 audit, dual-blind review filtered approximately 5-10% of LLM-generated mutants as semantically inconsistent; if these filtered mutants were systematically less effective at killing under MR_aligned (a plausible but unverified premise), the upper-bound δ-shift contribution from removing the reviewer step is bounded above by approximately **±0.03 to ±0.05 on Cliff's** δ at n_aligned = 12. This bound is an order of magnitude larger than the observed −0.009 source-axis shift and therefore prevents reading the source-axis null as evidence of LLM-source-diversity inertness; a direct dual-blind v4 rerun (deferred to a follow-up study) would give a tighter quantification.

0.8. 8. Conclusion

0.8.1. 8.1 Findings summary

The 60-cell empirical demonstration produces five findings.

(i) The H2 large-effect threshold is **not met under the pre-registered point-estimate criterion** (δ_v3 = 0.323; observed effect lies in the medium-effect range of Romano 2006). The 49.1% stipulated power at the large-effect boundary clarifies that "not met" is a statement about the point estimate, not about the effect size; we frame the finding as an underpowered exploratory contribution and defer full verification to a follow-up study with a larger PUT sample.

(ii) Holding the c-class primary metamorphic relation at the pre-registered MP5, three-LLM cross-source pooling under an identical prompt shifts Cliff's δ by only −0.009 (95% CI covers zero); within this design, **LLM identity under an identical prompt is not the lever** on the aligned-vs-cross effect size. A strong-sense source-diversity test with per-LLM differential prompts is deferred to a follow-up study.

(iii) Cross-source pooling raises mutant *quality* (mean C1_share +27%, class-c mean SMS +91.5%, class-d mean SMS +38.3%) without raising the aligned-vs-cross effect size. Source diversity therefore improves the kill set at the LRCA-classification level without expanding the operator-MP coverage that drives δ.

(iv) The Friedman main effect on MP differentiation is χ^2 = 15.30, p = 0.0041.

(v) The Spearman correlation between SMS and Pattern Coverage is 0.163 at n = 12. Orthogonality is a hypothesis, not a finding.

0.8.2. 8.2 Methodological contributions

This paper contributes a three-layer framework for domain-semantic mutation in single-output scientific computing kernels.

- **Layer 1 (§3.2).** Formal necessary conditions for semantic mutation, cross-function-boundary substitution, dependence on domain knowledge, change in algorithmic class, instantiated as five meta-operator classes: CE, OS, HP, TF, and SI (also written mut_C, mut_M, mut_G, mut_T, mut_F).
- **Layer 2 (§2.3).** The E1 $\wedge$ E2 equivalence judgement, the conservative complete instantiation of the Layer-1 conditions, with an explicit trade-off against E1-alone and E2-alone variants.
- **Layer 3 (§3.5).** AST-normalised empirical traceability across all 12 PUTs: a 5.14% overall AST overlap with cosmic-ray defaults, and HP, SI, and TF unreachable at 0/0/0 under default first-order configurations (HOM not refuted; see §3.6(ii)).

SMS is backward-compatible with the classical MS through §2.6 Theorem 9.1, which establishes almost-everywhere degeneration in the limit `L = L_equiv` $\wedge$ `L_killed` $\wedge$ `L_mut`. The 60-cell empirical audit reported in §5 is one demonstration following this backbone, not the paper's main contribution.

0.8.3. 8.3 Future work and P-series roadmap

We commit six P4 follow-ups.

(a) A pre-registered c-class primary-MP rule on a fresh dataset.

(b) A differential-prompt LLM-diversity test using V_canonical, V_persona, and V_cot (full protocol in Appendix C.1.1).

(c) A scaling study at $n \geq 30$ PUTs for SMS-vs-Pattern-Coverage orthogonality and for HOM-equivalence empirics.

(d) A rerun of the dual-blind reviewer protocol on the full v4 grid, to separate protocol asymmetry (R13) from source diversity.

(e) Cross-language portability work (Python $\rightarrow$ JavaScript and Julia) building on Tip, Bell, and Schäfer (2024).
(f) A self-hosted open-weight LLM pilot for air-gapped industrial deployment (P5 scope).

The companion P-series roadmap is as follows.

- **P1.** MR meta-pattern audit on the same 12-PUT infrastructure (under review at *Progress in Nuclear Energy* and *International Conference on Software Analysis, Evolution and Reengineering* (SANER) 2027).
- **P3.** Industrial-scale Java and C++ port with a second-rater inter-rater κ for LRCA.
- **P4.** Formal theorems on minimal MR-subset existence, reachable adequacy, and three-pillar coupling, targeted at *ACM Transactions on Software Engineering and Methodology*.
- **P5.** Regulatory transfer to IEC 60880, ISO 26262, and DO-178C with the conceptual complementarity argument (Chinese, in submission to *Nuclear Power Engineering*).

0.9. Data and code availability

All raw data, JSON SSOTs (paper_numbers_v3 / v4, rq2_cliffs_delta_v3 / v4_mp5, lrca_60cell, lrca_v4_mp5_recompute, rq2_power_stipulated, cosmic_ray_12put_ast_diff), mutant pools, AVP source, and analysis scripts will be archived on Zenodo at the time of acceptance under DOI [10.5281/zenodo.XXXXXXX] (placeholder, to be minted on acceptance). For peer review, an anonymized read-only mirror of the repository is available at [https://anonymous.4open.science/r/p2-sms-anon-XXXX] (URL to be provided by the corresponding author upon Editor request, per IST guidelines). The repository structure follows `REPRODUCIBILITY.md` in the source tree. The cosmic-ray and mutmut operator versions referenced in §3.5, §3.6, and Appendix B.6 are pinned in `requirements-frozen.txt`.

0.10. References

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Appendix .1. A. Notation and Operator Catalogue

Appendix .1.1. A.1 Complete notation table

```
═══════════════ Paper Domain ═══════════════
Paper identity:   P1, P2, P4, P5

═══════════════ Experimental Objects ═══════════════
PUT set I:        {A1,A2,A3, B1,B2,B3, C1,C2,C3, D1,D2,D3}, |I| = 12
PUT:              S_i  (i ∈ I)
Class mapping:    cls : I → {A, B, C, D}     (open, extensible)
Valid input distribution:    D_S, sampling X_{K_eq} ~ D_S

═══════════════ Metamorphic Testing Side ═══════════════
Meta-Pattern (MP):
  MP = {MP_1, ..., MP_5}, k ∈ {1,...,5}      (open, extensible)
  MP_1 Conservation  MP_2 Monotonicity  MP_3 Convergence
  MP_4 Trajectory    MP_5 Partial-order
Metamorphic Relation:  MR_{i,k} (provided by P1), mr = (r, R) ∈ MR_{i,
Automated Verification Pipeline (AVP):
  AVP : Programs × MR_universe × R⁺ → {pass, fail}
  AVP(s, mr, ε_AVP^k)  invokes the verification method per MP_k

═══════════════ Mutation Testing Side ═══════════════
Mutation Operator family (MUT):
  MUT = {mut_1, ..., mut_5}, j ∈ {1,...,5}    (open, extensible)
  mut_1 = mut_C  mut_2 = mut_M  mut_3 = mut_G  mut_4 = mut_T  mut_5 =
```

```
  Signature:  mut_j : Programs → 2^Programs
Mutant:        s' ∈ mut_j(S_i)
Alignment:     align(j) = j

═══════════════ Three-State Decomposition (Fixed) ═══════════════
mut_j(S_i) = equiv_{i,k,j} ∪ killed_{i,k,j} ∪ survive_{i,k,j}    (disj

equiv determination (dual conditions):
  (E1) AVP-coherent:  ∀ mr ∈ MR_{i,k}: AVP(S_i, mr) = AVP(s', mr)
  (E2) Output-equiv:  ∀ x ∈ X_{K_eq} ~ D_S: ||S_i(x) − s'(x)|| ≤ ε_eq

killed determination:
  killed(s', MR_{i,k}) ⇔ ∃ mr ∈ MR_{i,k}: AVP(S_i, mr) = pass ∧ AVP(s'

═══════════════ Metric (Fixed Classical Structure) ═══════════════
SMS_{i,k,j} := |killed_{i,k,j}| / (|mut_j(S_i)| − |equiv_{i,k,j}|)
            =  |killed_{i,k,j}| / (|killed_{i,k,j}| + |survive_{i,k,j}
            ∈ [0, 1]

═══════════════ LRCA Engineering Attribution Layer (Descriptive, Not i
Likely root-cause inventory:  C = {C1, ..., C5}      (open, extensible
  C1 True semantic failure / C2 Tolerance perturbation / C3 OOD
  C4 Statistical-assumption violation / C5 Mutator artefact
LRCA output:    each s' ∈ killed annotated with root_cause(s') ∈ C
Descriptive quantities:    C1_share_{i,k,j}, suspect_share_{i,k,j}
```

```
═══════════════ Slicing and Cross-Class ═══════════════
Slicing:         aligned: j = k    cross: j ≠ k
Cross-class:     ΔSMS_c, c ∈ {A, B, C, D}
                 CV(ΔSMS) := std(ΔSMS_c) / |mean(ΔSMS_c)|

═══════════════ Tolerance and Sampling ═══════════════
Tolerance:       ε_AVP^k (MP-dependent), ε_eq (equivalence output tole
Sampling:        K_eq = 1000 (E2 sample size), N = 20 (statistical rep
```

The notation skeleton (11 core concepts + three-state decomposition + SMS formula + AVP interface) is **fixed**; the *content* of MUT, MP, C, and cls is open to extension.

Appendix .1.2. A.2 LRCA decision tree and three-layer diagnostic protocol

Five likely root causes (open):

Code	Meaning
C1	True semantic failure (what P2 seeks)
C2	Numerical tolerance perturbation
C3	Out-of-distribution (OOD)
C4	Statistical-assumption violation
C5	Mutator artefact

Three-layer diagnosis with L0 artefact pre-scan:

```
L0 artefact pre-scan:    double-blind review (§4.2.4 dual-LLM cross-sc
```

```
L1 tolerance robustness: N = 20 replicates, fail ratio ≥ 0.80 consider
L2 OOD triage:           for C/D classes, distinguish valid D_S^valid
L3 statistical baseline: for Wilcoxon/DTW, IID/stationarity pre-check
```

Decision tree:

```
For each s' ∈ killed_{i,k,j}:
  L1: fail ratio < 0.80 → C2
       otherwise → L2
  L2 (C/D classes): fail only in OOD → C3
       otherwise → L3
  L3 (B/D classes + Wilcoxon/DTW): assumption violation → C4
       otherwise → artefact recheck
  Artefact evidence → C5
       otherwise → C1
```

Multi-label priority: C5 > C4 > C3 > C2 > C1.

LRCA threshold calibration (9-grid scan; `lrca_calibration.json`). Best ood_band=0.02 with any tolerance_multiplier (3.0 / 10.0 / 30.0) yields mean_C1_share 0.200 / H5 12/60 (20.0%); ood_band=0.05 (default) or 0.10 yields 0.164 / 10/60 (16.7%). `tolerance_multiplier` has zero impact (L1 tolerance rarely triggered); `ood_band` is the only discriminator. Calibration ceiling 12/60 = 20% remains far below 80% — H5 unattainable on this dataset, not a calibration issue (inherent SNR of LLM-mutant pools).

Appendix .1.3. A.3 AVP protocol specification and equivalence-judgement trade-off

`AVP : Programs × MR_universe × R⁺ → {pass, fail}`. Per-MP verification: MP_1 tolerance equality $|\mathrm{LHS} - \mathrm{RHS}| \leq \varepsilon$; MP_2 / MP_5 Wilcoxon signed-rank α=0.05; MP_3 convergence-order + asymptotic residual ratio; MP_4 DTW distance threshold ε_DTW. AVP version = P1 arXiv commit hash; P2 package embeds AVP source.

Three-candidate equivalence trade-off.

Determination	False-positive	False-negative	SMS bias
E1 alone	numerical coincidence + AVP-coherent	E1 false → non-equiv	Low (smaller denominator)
E2 alone	output match, MR differs (rare)	K_eq miss, full-space equal	High
E1 $\wedge$ **E2**	both fail simultaneously	E1 $\vee$ E2 false → non-equiv	Slightly high (conservative)

E1 alone is fooled by insufficient AVP coverage; E2 alone by numerical coincidence on K_eq. E1 $\wedge$ E2 is the conservative complete instantiation; under `L_equiv` it reduces almost-everywhere to classical bitwise equivalence (Lemma 9.1). Counter-examples: E2 passes but E1 does not (rare; mutant coincides at K_eq but deviates at MR trigger points → non-equivalent); E1 passes but E2 does not (common; consistent within MR but numerical drift > ε_eq → non-equivalent).

Appendix .2. B. Experimental Subjects and Mutation-Operator Specialisations

Appendix .2.1. B.1 PUT selection rationale (detailed coverage argument)

(a) Library-stack coverage. numpy 2.4.4 (A1-B3 linear algebra / arrays), scipy 1.17.1 (A1 integrate, A2 linalg, B2 stats), scikit-learn 1.8.0 (C1-D3 surrogate / ML) — Python scientific-computing's de facto foundation stack (PyPI top-50, 2026-04). Uncovered: GPU / distributed (JAX, CuPy, dask), domain-specific (BioPython, Astropy, RDKit). Reserved for P3 (R5).

(b) Mathematical-structure coverage. ODE/PDE (A1, A3), direct linear algebra (A2), Bayesian analytic + MCMC + Monte Carlo (B1-B3), kernel + orthogonal polynomial + NN surrogate (C1-C3), convex optimisation + backprop + max-likelihood (D1-D3) — covers 8 of 12 chapters in Numerical Recipes (Press et al. 2007). Uncovered: advanced PDE solvers (FEM, FV, spectral); FFT; interior-point/trust-region optimisation; symbolic/CAS. Reserved for P3.

(c) Comparison with benchmarks. DeepCrime (Humbatova et al. 2021): P2 class D topical overlap on ML kernels. Defects4J (Just et al. 2014): no overlap, only mutation-as-fault-proxy reference. mutmut / cosmic-ray demos: general Python, P2 extends scientific-computing focus.

(d) Scale vs representativeness. Each PUT 50-400 LOC; signature standardised to `program(x: float)` $\rightarrow$ `float`. Substantively smaller than industrial code (typically 1-10 KLOC).

Limitation. The `float` $\rightarrow$ `float` signature is a substantive constraint, not engineering trade-off. Industrial inputs are typically high-dimensional (CFD grids, MD particles, FEM matrices); scalarised PUTs upper-bound mutant semantic complexity and may systematically under-estimate SMS and cross-class differences on industrial PUTs (declared §7 R5; P3 validates on industrial-grade PUTs).

Appendix .2.2. B.1.5 Systematic vs incidental

Syntactic tools may *occasionally* hit §3.2 (a) (cross-function-boundary) or (c) (algorithm class), but occasionality undermines two engineering functions of semantic mutation. **(i) Deepening source-code understanding.** Designing OS `np.linalg.det(M)` $\rightarrow$ `np.sum(np.diag(M))` requires knowing the two are equivalent on diagonal matrices but not on general matrices (`det =` $\prod$ `eigvals` vs `sum(diag) = trace =` $\sum$ `eigvals`); syntactic tool AST traversal does not require this understanding even when similar replacements appear. **(ii) Revealing deep faults.** Domain-semantic errors (physical-constant, unit conversion, boundary conditions, hyperparameter semantics, numerical-method order) are not AST-local; syntactic mutator design goals (operator typos, off-by-one, negation flips) have hit probability for domain errors much smaller than systematic semantic-mutator triggering, and lack repeatability. **Conclusion.** Syntactic tools occasionally producing mutants satisfying (a)(b)(c) is stochastic byproduct — not repeatable, not carrying engineering value. Systematic semantic mutation requires (a)(b)(c) to be design intent.

Appendix .2.3. B.2 Per-class operator specialisations

Operator	Representative specialisations across PUTs
mut_C Conservation-breaking	A1 Lorenz: add ε_drift to RHS (slow Hamiltonian drift); A2 LU: decomposition omits k+1-th multiplier (breaks det conservation); B1 Beta-Bin: posterior omits normalisation; C1 GPR: covariance omits positive-definite diagonal term; D1 MLP: backprop omits one gradient term.
mut_M Monotonicity-breaking	A3 FDM: Δt coefficient occasionally negative; B2 MCMC: acceptance min(1, r) $\rightarrow$ min(0.95, r); C2 PCE: high-order coefficient sort inserts inversion; D2 SVM: decision-function sign flips near boundary.
mut_G Convergence-breaking	A1 Lorenz: RK4 $\rightarrow$ 1.5-order hybrid; A3 FDM: 2nd-order difference $\rightarrow$ 1st-order; B3 MC: doubling sample size does not 1/N-reduce variance; C3 NN-Surr: training-epoch truncation.

Operator	Representative specialisations across PUTs
mut_T Trajectory-distorting	A1 Lorenz: state-vector y/z swap; B2 MCMC: insert independent-sampling segment; C3 NN-Surr: training-target slow phase shift; D1 MLP: hidden-layer periodic-mask activation.
mut_F Fidelity-order-breaking	A2 LU: partial pivoting degrades to no pivoting; C1 GPR: length-scale switches to coarse prior; C2 PCE: high-order term randomly retains low-order; D3 LR: regularisation occasionally large.

Per-class HP / OS / TF substitution rules (illustrative). **HP** on class a: tolerance / max_iter; on class c: GPR `noise_level` / `length_scale`; on class d: MLP `hidden_dim` / dropout. **OS** on class a: numerical-linalg API swaps (`det` $\leftrightarrow$ `sum(diag)`); on class b: probability-distribution sampling swaps; on class c: surrogate-class swaps (GPR $\leftrightarrow$ RBF $\leftrightarrow$ NN). **TF** on class a: integration-order changes (RK4 $\rightarrow$ Euler); on class b: MC estimator changes.

Operator-level cross-table with cosmic-ray defaults (R-15 response). The 12 cosmic-ray default operators are AST-local: `NumberReplacer` (Num/Constant — CE partial, △ literal values only); `ReplaceArithmeticOperator` (BinOp); `ReplaceComparisonOperator` (Compare); `ReplaceLogicalOperator` (BoolOp); `ReplaceUnaryOperator` (UnaryOp); `ReplaceTrueFalse` (Name-

Constant — CE keyword, △ Bool literal); `BreakContinueReplacer` (Break/Continue); `RemoveDecorator`; `RemoveExceptHandler`; `ZeroIterationForLoop` (For); `ReplaceIfBlock` (If); `MutateSubscript`. None correspond to OS / HP / TF / SI:

P2 class	Tool support	Coverage
OS API replacement	None	△ 88.33% disjoint (incidental low-complexity hits)
HP	None	× tool inexpressible (0/72)
TF numerical-method order	None	× tool inexpressible (0/54)
SI / CF structural injection	None	× tool inexpressible (0/33)

Operator-level conclusion. All 12 default classes remain AST-local (BinOp / Compare / BoolOp / UnaryOp / NameConstant / Subscript / If / For / Break / Continue / decorator / except). Categorically, no entry recognises sklearn / scipy hyperparameter semantics (HP), numerical-method order (TF), or control-flow intent (SI / CF). For OS, low-complexity sub-expressions can be incidentally hit by BinOp; empirical 11.67% (§3.5) is consistent with 88.33% AST-disjointness lower bound.

Appendix .2.4. B.3 cosmic-ray a1 single-PUT pilot (pre-12-PUT empirical)

Before the 12-PUT generalisation (§3.5 main), a single-PUT lightweight empirical ran via `scripts/run_cosmic_ray_a1.sh` on a1 (file 934 B, AST-bound). Outcome: mutants_generated ~tens; $\geq$ 90% belong to BinOp / Compare / 12 default classes; small killed-by-`test_a1.py` ratio (P2 unit tests are output-shape/type sanity checks; most cosmic-ray mutants not killed) — empirical manifestation of the §3.2.6 thesis that syntactic-tool mutants are not aligned with MR-violation detection. The 12-PUT generalisation (§3.5) supersedes this pilot but corroborates the same conclusion at scale.

Appendix .2.5. B.4 Expected LRCA risk profile

PUT-class $\times$ LRCA-layer risk weights. A numeric: C2 dominant ($\star\star\star$ on L1 tolerance). B probabilistic: C4 dominant ($\star\star\star$ on L3). C surrogate: C3 dominant ($\star\star\star$ on L2 OOD; $\star\star$ tolerance). D ML: C3 + C4 mixed ($\star\star\star$ L2; $\star\star$ L3).

Operator-PUT root-cause hotspots (expected). A: mut_C/G/F dominantly C2 (numerical tolerance), mut_M/T C1 (true semantic). B: dominantly C4 (statistical-assumption violations). C: mut_C/M dominantly C2/C1 on c1/c2, mut_T/F dominantly C3 on c1/c2; c3 (NN-Surr) row dominantly C5 (artefact). D: d1 row dominantly C5 (training-noise artefact); d2/d3 dominantly C1 / C3 mixed.

Expected suspect_share thresholds. A: 0.10-0.20 (acceptance $\leq$ 0.25); B: 0.20-0.35 ($\leq$ 0.40); C: 0.20-0.30 ($\leq$ 0.35); D: 0.25-0.40 ($\leq$ 0.45). 60-cell average: $\leq$ 0.20 (= H5; acceptance $\leq$ 0.25).

Appendix .2.6. B.5 Engineering-significance mapping

j = k aligned diagonal is the H2 threshold-test slice; j $\neq$ k off-diagonal is control; 6 vacant cells ($\circ$, P1 H6) are not formally adjudicated and are repurposed in

§6.2 as descriptive evidence for R_sem / R_kill decoupling. The v3 → v4 micro-change (−0.009) attributes to LLM-source diversity under a fixed prompt; this is the source-axis null contrast underpinning §5.3.

Appendix .2.7. B.6 Mutmut vs cosmic-ray default-operator overlap

Reviewer-requested manual operator-class cross-reference between cosmic-ray's 13 default operators (cosmic-ray 8.4.6, `cosmic_ray.operators` package) and mutmut's 14 default operators (mutmut current `main`, `src/mutmut/mutation/mutators` Both default-operator sets are first-order AST-local replacements; neither implements a cross-function-boundary, domain-knowledge-dependent, or algorithmic-class-changing mutation (the three §3.2 necessary conditions for semantic mutation).

Operator class	cosmic-ray default	mutmut default	Reaches §3.2 (a/b/c)?	P2 class reachability
Numeric literal ±1	`number_replacer`	`operator_number`	(a) no, (b) no, (c) no	partial CE only (e.g. `_RHO=28.0` → `27.5`)
Binary arithmetic swap (+, −, ×, ÷)	`binary_operator_replacement`	`operator_swap_op` (binary subset)	(a) no, (b) no, (c) no	partial OS only (incidental)
Comparison swap (<, ≤, >, ≥, ==, !=)	`comparison_operator_replacement`	`operator_swap_op` (compare subset)	(a) no, (b) no, (c) no	none

Operator class	cosmic-ray default	mutmut default	Reaches §3.2 (a/b/c)?	P2 class reachability
Boolean swap (and ↔ or)	`boolean_replacer`	`operator_swap_op` (bool subset)	(a) no, (b) no, (c) no	none
Unary remove / swap	`unary_operator_replacement`	`operator_unary_ops`, `operator_swap_op`	(a) no, (b) no, (c) no	none
Keyword swap (True/False, etc.)	`keyword_replacer`	`operator_keywords`, `operator_name`	(a) no, (b) no, (c) no	none
break ↔ continue / return	`break_continue`	`operator_keywords` (subset)	(a) no, (b) no, (c) no	none
Exception class swap	`exception_replacer`	(none)	(a) no, (b) no, (c) no	none
Decorator removal	`remove_decorator`	(none)	(a) no, (b) no, (c) no	none
Variable-binding insert / replace	`variable_inserter`, `variable_replacer`	`operator_arg_removal`, `operator_assignment`	(a) no, (b) no, (c) no	none
no_op insertion	`no_op`	(none)	(a) no, (b) no, (c) no	none
Zero-iteration `for` loop	`zero_iteration_for_loop`	(none)	(a) no, (b) no, (c) no	none

Operator class	cosmic-ray default	mutmut default	Reaches §3.2 (a/b/c)?	P2 class reachability
String content tweak (XX prefix, case flip)	(none)	`operator_string`	(a) no, (b) no, (c) no	none
Lambda body → `None`/`0`	(none)	`operator_lambda`	(a) no, (b) no, (c) no	none
Dict kwarg name (XX prefix)	(none)	`operator_dict_arguments`	(a) no, (b) no, (c) no	none
Symmetric/asymmetric string-method swap	(none)	`operator_symmetric_string_methods_swap`, `operator_unsymmetrical_string_methods_swap`	(a) no, (b) no, (c) no	none
Augmented → simple assignment	(none)	`operator_augmented_assignment`	(a) no, (b) no, (c) no	none
`match` case removal	(none)	`operator_match`	(a) no, (b) no, (c) no	none

Aggregate. Cosmic-ray ∩ mutmut: ~9 of 13 cosmic-ray operators have a near-equivalent in mutmut (numeric, binary, compare, boolean, unary, keyword, break-continue, variable, plus partial overlap on string-prefix vs `XX`); 4 cosmic-ray operators are unique (exception, decorator, no_op, zero-iteration-for); 6 mutmut operators are unique (string content, lambda, dict-kwarg, string-method swap, aug-assignment, match-case). The union is approximately 21 distinct first-order AST-local operator classes. None of the 21 can:

- (a) Cross a function-call or module-import boundary (e.g. replace `det(M)` with `sum(np.diag(M))` requires knowing the equivalence on diagonal matrices — outside any default-operator scope);
- (b) Depend on domain knowledge for legality (e.g. a hyperparameter swap that knows `noise_level=1e-4` is the load-bearing knob in a Gaussian Process Regression PUT requires PUT-class awareness);
- (c) Change the algorithmic class (e.g. swap `polynomial` → `spline basis` rewrites the surrogate's mathematical structure).

The mutmut / cosmic-ray null intersection on §3.5's HP/SI/TF classes (zero overlap) therefore reflects a **structural property of first-order AST-local mutation** rather than a tool-specific limitation, supporting the §3.6 preventive-defence framing.

Appendix .3. C. Experimental Procedure Details

Appendix .3.1. C.1 Cross-source v4 protocol — full specification

Two-stage ablation.

Version	Mutant pool	c-class primary MP	Use
v3	Single-source (Claude Opus 4.6)	MP5 (P1 legacy, pre-registered)	H2 baseline
v4	**Cross-source** (Claude Opus 4.6 + GPT-5.4 + DeepSeek chat)	MP5 (held at pre-registered)	Isolate LLM-source diversity

Protocol (`scripts/cross_source_campaign.py`). (a) For each (PUT, operator), Claude / GPT / DeepSeek run K = 3 trials with identical prompt (§4.2.2), temperature 0.7; `source_tag` propagated to filenames. (b) **V1-V4 mechanical validation** (`src/p2/mutators/validation.py`): V1 syntax (`ast.parse`), V2 executability, V3 non-triviality (`|y_mutant - y_original| > 1e-6`), V4 signature consistency. (c) **DeepSeek model:** `deepseek-chat` (113 tokens/call, 1.8 s), not `deepseek-v4-pro` (340 tokens/call, 11 s) — quality-equivalent on a2_OS1 dry-run. (d) **Pool capacity:** 37 ops $\times$ 3 sources $\times$ 3 trials = 333 attempts; V1-V4 pass rate 89%; 298 confirmed mutants. Sources contribute nearly equally: Claude 101 / GPT 98 / DeepSeek 99 (Phase A engineering finding). (e) **Sampling** (`scripts/build_pools.py`, POOL_VERSION = v4): per-PUT 30 max, measured mean 24.3, range 10-30. c1 (GPR) yields only 10 because c1_HP1 / c1_CE1 have V1-V4 near-zero pass (WhiteKernel noise_level 1e-4 $\rightarrow$ 1e-1 perturbation has minimal output impact, triggers V3 non-trivial failure — itself §6.2 evidence).

Protocol-asymmetry (R13). v4 does not invoke reviewer LLM (cost / speed priority; a follow-up study will rerun dual-blind on the v4 grid, ~$5-8 USD); v3 used the original Phase-1 dual-blind (Claude gen + GPT-5.4 review + DeepSeek arbitration). A fraction of the v3 $\rightarrow$ v4 source-axis shift may be slight v4 quality decline rather than LLM-source diversity.

Appendix .3.2. C.1.1 Differential prompt protocol (future-work commitment)

The v3 $\rightarrow$ v4 micro-change of −0.009 may reflect "three LLMs under identical prompt" rather than "true upper bound of LLM diversity". Separation requires *one tailored differential prompt per LLM* (skeleton: `scripts/run_differential_prompt.py`).

Design (2×3 factorial, within-(PUT, operator)). Factor A — prompt template, 3 levels: **V_canonical** (control, identical to v4), **V_persona** (per-LLM identity: Claude "numerical-analysis editor"; GPT "scientific-software refactorer"; DeepSeek "library-API substitution specialist"), **V_cot** (Claude <thinking> tags; GPT "step by step"; DeepSeek stepped-reasoning). Factor B — LLM source (Claude / GPT-5.4 / DeepSeek chat). K = 3 per cell; total 37 $\times$ 3 $\times$ 3 $\times$ 3 = **999 trials**.

Exit criteria. If max(delta) − min(delta) < 0.05: the v3 $\rightarrow$ v4 −0.009 source-axis null is robust. If $\geq$ 0.05 with prompt variant pushing delta past 0.474: H2 revised from "not met" to "prompt-conditional met". If $\geq$ 0.05 but all delta < 0.474: prompt sensitivity exists but H2 unchanged.

Resources. API ~$18-30 USD; wall 3-4 h at concurrency 4; V1-V4 only, no reviewer LLM.

Appendix .3.3. C.2 Manual pilot lineage, LLM-client configuration, and L0 prescreen

The original §4.2 protocol used 60% multi-LLM consensus (Claude Opus 4.6 + GPT-5.4 + DeepSeek chat unanimous) plus 40% manual injection by scientific-computing researchers. Each (mut_j, S_i) pair carried one prompt template containing PUT source, semantic intent of mut_j, prohibition against revealing specific MRs (avoids prompt leakage), and output requirements (syntactically correct, executable, single-point modification, diff < 10 lines). Reproducibility parameters: temperature 0.3 in manual-pilot stage (v4 cross-source uses 0.7 per §C.1); fixed seed; 5 candidates per prompt with 2-3 retained.

Double-blind review (Scheme C). Generator LLM-G (Claude Opus) and reviewer LLM-R (GPT-4o) are cross-source with strict role separation. LLM-R sees only PUT original + mutant code, unaware of mut_j category / MR content / generator identity, outputting a triple (syntactic, executable, semantic-fault-injected). Double-confirmed mutants enter pool; inconsistencies enter manual arbitration ($\leq$ 10%). From the double-confirmed pool, 20% is sampled for manual review; manual-vs-LLM-R inconsistency > 10% triggers full manual downgrade. Same-source bias mitigation stratifies LLM-R by PUT class. Prompt-injection mitigation disables RAG / web search on the API path.

L0 pool prescreen. Each mutant passes static syntax (linters / type checkers), a unit self-test (simple inputs produce finite output), and double-blind sign-off. Target: 30-50 candidates per cell $\rightarrow$ 10-15 retained after prescreen.

Appendix .3.4. C.4 LRCA three-layer execution (full)

Decision tree, multi-label priority (C5 > C4 > C3 > C2 > C1), and output (`C1_share`, `suspect_share`) are stated in §A.2; the full pseudocode is reproduced there. LRCA does **not** modify SMS; killed set is not filtered. §5.7 H5 numbers use best calibrated combination (`ood_band = 0.02, tolerance_multiplier = 3.0, repeats = 20`); default-threshold results retained in `lrca_60cell_v3.json` as control. Interface with §3.6 risk profile: L0 prescan $\leftrightarrow$ §C.2 dual-LLM double-blind; L1 tolerance $\leftrightarrow$ N = 20 repetition subprocess; L2 OOD $\leftrightarrow$ input-distribution definition; L3 statistical baseline $\leftrightarrow$ pre-check protocol.

Appendix .3.5. C.4.1 Pilot calibration (37 operators, K = 10/20)

A 2026 Q3 operator-level pilot ran 12 PUTs $\times$ 37 operators (12 `is_key=True` at K=20, 25 at K=10; 470 trials total). Pilot precursor quantities: R_sem (V1-V6 $\wedge$ operator_match pass rate), D_impl (median pairwise AST + literal + identifier Jaccard distance among confirmed mutants), R_kill (proportion of confirmed mutants killed by AVP under that PUT's primary MP). Aggregate (N=37): R_sem median 0.50 mean 0.468 (0/37 with R_sem=0); D_impl median 0.42 mean 0.392 (1/37 with D_impl $\approx$ 0); R_kill median 0.00 mean 0.189. All 37 operators produced $\geq$ 1 V1-V6 passing mutant (verifying H1).

Key R_sem / R_kill decoupling pattern. R_sem $\geq$ 0.9 $\wedge$ R_kill = 0 combinations all on HP / TF / OS operators in c / d classes (e.g., c1_CE1 noise 1e-4$\rightarrow$1e-1 with R_sem 1.00, R_kill 0.00; c3_HP1 relu$\rightarrow$tanh, c3_TF1 max_iter 1000$\rightarrow$5, d1_HP1 MLP α, d3_HP1 LR C). R_sem-high $\wedge$ R_kill = 1 combinations all on CE / OS operators in a / b classes (a2_CE1 LU det, a2_OS1 prod$\rightarrow$sum, b1_OS1 α/β swap, d1_TF1 label flip). This is empirical evidence for §6.2 R_sem / R_kill decoupling at cell level.

Appendix .4. D. Statistical Analysis Details

Appendix .4.1. D.1 SMS variants and construction lemmas

SMS_unfiltered (reviewer comparison metric):

```
SMS_unfiltered_{i,k,j} := |killed_{i,k,j}| / (|mut_j(S_i)| − |equiv_{i
                          (same form as primary; killed does not disti
```

The appendix of the reproducibility package provides cell-by-cell difference between SMS_unfiltered and SMS; if relative difference < 5%, this confirms LRCA does not affect robustness of primary conclusions.

Primary-table construction.

RQ	Primary statistics	Reporting format
RQ1	inst_rate, equiv_rate, C1_share, survive_rate	60-cell heatmap + 4-class marginals
RQ2	aligned-SMS vs cross-SMS; sparse ○ vs dense ••equiv_rate	Cliff's delta + odds ratio + 95% bootstrap CI
RQ3	ΔSMS_c (c ∈ {A,B,C,D}); CV(ΔSMS)	sign test (df=3) + descriptive forest plot
RQ4	Spearman ρ + Kendall τ for SMS vs PC	scatter + dual-metric ranking comparison

Multiple comparisons. Cell-level claims across 60 cells use Benjamini-Hochberg FDR, alpha_FDR = 0.05. N = 20 bootstrap intervals: 1,000-iteration 95% CI (B = 10,000 for the headline H2 delta CI).

Appendix .4.2. D.2 Cliff's delta cutoff sensitivity and logit-transformation invariance

H5 cutoff sensitivity (R-14 response). Dense grid scan (cutoff $\in$ {0.05, 0.10, …, 0.50}, step 0.05) on v4 data (`scripts/h5_sensitivity.py`, `data/results/h5_sensiti`

cutoff	h5_cells_pass	h5_pass_ratio
0.05	12 / 60	20.0%
0.10	12 / 60	20.0%
0.15	12 / 60	20.0%
0.20 (paper)	**12 / 60**	**20.0%**
0.25	12 / 60	20.0%
0.30	12 / 60	20.0%
0.35	12 / 60	20.0%
0.40	12 / 60	20.0%
0.45	13 / 60	21.7%
0.50	13 / 60	21.7%

Conclusion. H5 verdict (not met) is intrinsic data property, independent of cutoff choice. v4 suspect_share distribution is severely bimodal: median 1.0, mean 0.79; 48 cross cells at suspect_share $\approx$ 1, 12 aligned cells in low end; the [0.20, 0.80] interior is nearly empty. Specific value 0.20 is **not load-bearing**.

Logit-transformation invariance. Cliff's delta is rank-based (function of U / (n1·n2)), mathematically invariant under any strictly monotone transformation (Romano 2006). Logit is strictly monotone on (0, 1), so delta_logit ≡ delta_raw is a construction result. **Not additional robustness evidence**; H2 robustness threats come from §5.2 zero-mass dominance, not metric-scale choice.

Appendix .4.3. D.3 Power analysis — plug-in and stipulated-alternative

Plug-in bootstrap. With-replacement sample from observed aligned (n=12) and cross (n=48) v4 SMS pools; N_sim = 5,000, seed = 42 (`scripts/compute_rq2_power.py`, `rq2_power_v4.json`):

Threshold	Interpretation	Power
delta > 0.000	Any-effect	**0.997**
delta > 0.147	Small	0.966
delta > 0.330	Medium	0.759
delta > 0.474	**Large (H2)**	**0.423**

Sample-size sweep (n_aligned $\in$ {6, 12, …, 60}, n_cross = 4$\times$): power for delta > 0 reaches 0.974 at n=6, 0.996 at 12, plateaus thereafter. RQ2 primary analysis has very low sample-size requirement; the bottleneck is the effect-size boundary itself, not sampling noise. Even at delta_truth $\approx$ 0.474, sample has only ~42% chance of detecting under plug-in; but this does **not** mean "insufficient power is the cause of H2 not being met" — observed delta_v3 = 0.323 < 0.474, observed delta_v4 = 0.314 < 0.474; increasing sample only narrows CI.

Stipulated-alternative (R1 W1 round-2). Implementation via mixture-weight (`scripts/compute_rq2_power_stipulated.py`, N_sim = 2,000): SMS has heavy ties at zero (45/60 = 0), raw shifting is discontinuous (any $\varepsilon > 0$ jumps delta from 0.314 to 0.74). Mixture: aligned' = (probability w) sample from (observed_aligned + 0.001) + (probability 1−w) sample from observed_aligned, calibrate w so E[delta] $\approx$ 0.474. Calibrated w = 0.094, realised E[delta] = 0.4746.

Stipulated truth	Criterion	Power
0.474	delta_hat $\geq$ 0.474 (point-estimate)	**0.491**
0.474	95% CI lower > 0 (any-effect)	0.868

Even at delta_truth = 0.474, the (12, 48) design returns delta_hat $\geq$ 0.474 in only ~49% of replications — this *supports* the §5.3 framing that "not met" is point-estimate, not effect-size.

Appendix .4.4. D.4 Friedman per-class — discussion of small-N concordance

Per-class Friedman (3 PUTs $\times$ 5 MPs) with Bonferroni $\times$ 4: a chi^2=4.00 raw 0.406 adj 1.000, W=0.333; b chi^2=10.78 raw **0.029** adj **0.116**, W=**0.898**; c chi^2=4.00 raw 0.406 adj 1.000, W=0.333; d chi^2=5.00 raw 0.287 adj 1.000, W=0.417. After correction, no per-class result remains significant at family-wise alpha = 0.05. Even b-class W = 0.898 (Cohen 1988 large concordance) is insufficient at N = 3 per class. **H4 verdict rests on §5.5 sign test**; per-class Friedman is sensitivity / descriptive only. Friedman main effect on full 60 cells (chi^2 = 15.30, p = 0.0041) confirms MP rank differences but is logically independent from H4 cross-class consistency.

Mixed-effects unavailability. Primary `sms ~ C(class) + C(operator) + C(class):C(operator) + (1 | put)` returned Singular (insufficient column rank for class $\times$ operator interaction; N=60 too small for 11-d fixed-effects + 12 PUT random intercepts). Fallback `sms ~ C(class) + C(operator) + (1 | put)` had PUT random-intercept variance hit boundary 0 (degenerates to OLS). Fallback class p-values (descriptive only): b vs a 0.275 / c vs a 0.892 / d vs a 0.991. RQ3 conclusion shifts to four-piece presentation (class means + sign test + Friedman + forest plot).

Appendix .4.5. D.5 Pattern-coverage operationalisation

Per PUT, (MP_k, R_outcome $\in$ {True, False}) binary-tuple coverage: 5 MPs $\times$ 2 outcomes = 10 cells, PC = #triggered / 10. Simplest baseline (per-PUT granularity, not distinguishing mutants). PC range over 12 PUTs: [0.500, 1.000], mean 0.733 (`paper_numbers_v4.json`). Pairing per PUT: Spearman rho = 0.163 (p = 0.613); Kendall tau = 0.136 (p = 0.568).

Power caveat. At n = 12, Spearman 95% CI $\approx$ [-0.5, +0.6]; cannot distinguish "zero correlation" from "moderate positive" or "moderate negative". p = 0.61 / 0.57 means "no correlation detected at n = 12", not "correlation does not exist".

Within-class descriptive. b2 PC = 1.0 with mean SMS = 0.067; b1 PC = 0.7 with mean SMS = 0.20. c3 PC = 1.0 with mean SMS = 0.14; c1 / c2 PC = 0.7-0.8 with mean SMS = 0.0. Within the same class, higher-PC PUT has lower SMS — contradicts naive "more PC kills more mutants" assumption. Suggestive of orthogonality but n = 12 cannot confirm. P4: expand to n $\geq$ 30 + refine PC to incorporate mutant dimension.

Appendix .5. E. Stakeholder Deployment Considerations (Single-Output Kernels Scope)

Appendix .5.1. E.1 Air-gap incompatibility — full justification

The §6.4 workflow depends on external LLM API calls (Claude / GPT / DeepSeek), incompatible with most regulated air-gapped V&V workflows:

Domain	Standard	Air-gap requirement
Nuclear	IEC 60880	Air-gapped build for safety-critical software
Aerospace	DO-178C	Tool-qualified offline build for DAL-A/B; LLM API outside qualified-tool envelope
Medical	IEC 62304	Class C software lifecycle requires offline reproducible build
Automotive	ISO 26262	ASIL-D requires offline tool chain; cloud LLM non-compliant

Mitigation paths (P5). (i) Self-hosted open-weight LLMs (Llama / DeepSeek local inference) — capability gap vs API-served frontier models; quantitative impact on SMS untested. (ii) Offline-cached pre-generated mutant pools with commit-hash + raw-response signature locking — adopters reproduce a published pool offline (this paper's package supports this), but generating new pools requires external LLM access. SMS for air-gapped industrial deployment is a P5 research direction.

Appendix .5.2. E.2 Quarterly batch audit workflow

The original §6.5.2 per-PR GitHub Actions YAML was removed because PR-CI threshold 0.10 contradicted §6.5.3 audit threshold 0.20 and propagated stakeholderfacing CI assumptions into adopters' pipelines.

Replacement: quarterly batch audit:

Step	Description	Cost
1	Offline mutant-pool generation (per PUT 24-30 × 12 PUTs ≈ 300)	LLM API \$5-15; ~30 min
2	`sms_campaign.py --track 2`	4-core laptop ~10-20 min
3	`run_lrca.py`	< 1 min
4	MR-design team review of MRs with aligned-cell SMS significantly below historical median (0.275)	1-2 h meeting

Quarterly: API ~\$10 + automation 30 min + manual 1-2 h ≈ 0.5 person-day. Affordable. Per-PR gating not recommended (LLM latency 5-30 s + cost + non-determinism + air-gap incompatibility); adopters needing per-PR should use coverage / unit-test gates. SMS does not replace domain-expert MR physical-reasonableness judgement nor product-requirements review.

Appendix .5.3. E.3 V&V documentation — conceptual complementarity with ASME V&V 20-2009

§6.5.3 was retitled from "Auditors / certification bodies" in R3 round-2: this subsection makes **no normative claim** toward NRC, FDA, or ISO 26262 review teams. Within single-output kernels scope, no traceable mapping exists to current

IEC 60880 / ISO 26262 / DO-178C / ASME V&V 20-2009 normative bodies.

V&V documentation for scientific computing software (per ASME V&V 20-2009 §3 and similar guides) requires quantifiable test-adequacy evidence; current documentation often relies on code coverage + MR lists + SME signatures. SMS may appear as **research-grade supplementary evidence** alongside coverage and MR lists: (a) aligned-cell SMS per critical PUT; (b) LRCA three-layer diagnosis (C1 readout / C2-C5 attribution); (c) 60-cell matrix visualisation. Reviewers can independently run `REPRODUCIBILITY.md`.

Original threshold recommendations (aligned-cell SMS $\geq$ 0.20 / 0.30 + C1_share $\leq$ 0.20) were removed: no normative backing in IEC / ISO / ASME; misreads as enforce-ready. **SMS reported as descriptive supplementary evidence**, with adequacy judgements left to V&V reviewers case-by-case.

Conceptually complementary to ASME V&V 20-2009 §3 code verification (numerical-solver correctness) vs SMS (MR-set fault-detection adequacy). Substantial scale gap to multi-module CFD codes; incorporation into V&V standards body would require P5 large-scale empirics + multi-year dialogue with ASME V&V 20 / IEEE 1012 / IEC 60880 committees. **No advocacy that SMS enter any normative certification system in 2027.** SMS does not replace SME signatures, FMECA, PIRT, system-level V&V, or ASME V&V 20 §3 numerical-solver verification — it is one link in the evidence chain, not a single-point qualification.

Appendix .6. F. Threats to Validity — Detailed Mitigation

Appendix .6.1. F.1 Internal threats

ID	Threat	Mitigation	Residual
R1	LLM generation reproducibility	§C.2 reproducibility parameters in package; dual-LLM cross-source + 20% manual; identical prompt+seed should yield ≥ 90% pool overlap	LLM training data may be updated post-submission
R2	Probabilistic equiv (K_eq=1000)	§2.3 declares probabilistic approximation; Hoeffding-style false-equiv bound	K_eq sweep ∈ {500,1000,2000} deferred to P4 (R1 W3 round-2)
R3	P1 ↔ P2 circular dependency	AVP version pinned to P1 commit hash; package embeds AVP source; §1.6 declares P1 as arXiv technical report	None within scope

ID	Threat	Mitigation	Residual
R4	LRCA multi-label boundary	§A.2 priority C5>C4>C3>C2>C1; multi-label co-occurrence table in package; root_cause is “likely root cause” not definitive	C2-C5 may multi-trigger on B/D classes
R9	Mutant-pool size	Pool expanded to mean 17.4: delta moved 0.321 → 0.323, CI narrowed; **effect size is intrinsic ceiling, not pool dilution**	Some PUTs < 30 (cache-bound)
R10	LLM non-determinism	Multi-turn de-dup; K=10/20 repetitions; raw prompt+response committed for direct reuse	Claude Opus subscription lacks seed control

ID	Threat	Mitigation	Residual
R13	**Protocol-implementation gap v3 vs v4**	v4 used 3 providers (Claude 101 / GPT 98 / DeepSeek 99) without dual-blind; v4 C1_share 0.209 > v3 0.164 weakly opposes "v4 quality decline"	Complete separation requires a follow-up dual-blind rerun on the v4 grid

R13 is the protocol-asymmetry threat (dual-blind vs V1-V4 only) on the v3 → v4 source-axis contrast ($\Delta\delta = -0.009$).

Appendix .6.2. F.2 External / construct / conclusion threats

ID	Threat	Class	Mitigation
R5	12-PUT representativeness	External	Follows P1 site selection; §1.6 P-II open principle (cls extensible); P3 scaling to MD/QC/CFD

ID	Threat	Class	Mitigation
R6	Cross-class statistical power	Conclusion	H4 framed exploratory; mixed-effects unavailable (Singular at N=60); shifted to class means + sign test + Friedman + forest plot
R7	LLM homogeneity bias	Construct	§C.2 three-LLM rotation + PUT-class subdivision + 20% manual sampling; third-party LLM family validation deferred

ID	Threat	Class	Mitigation
R8	LLM-source distributional shift	External	Hit distribution DeepSeek 11/15, Claude 4/15, GPT 0/15; §3.5 argument is categorical AST-locality, not hit frequency; HP/SI/TF zero-overlap across all 3 sources unaffected
R12	HOM equivalence	External	Tool-unreachability claim confined to first-order syntactic tools; HOM empirical testing listed as P4 future work

Conclusion threats: multiple comparisons handled by BH-FDR at alpha_FDR=0.05 (§D.1); N=20 stability handled by 1,000-iteration bootstrap CI (§D.3). Construct threat — SMS measures epistemological semantic detection, not production engineering value (P2-CN scope).

Appendix .7. G. SMS → MS Degeneration Theorem — Full Proof

Appendix .7.1. G.1 Notation cross-reference (with §2.1)

PUT `S_i`, mutation operator family mut (syntactic or semantic), mutant set mut(S_i), MP set, equivalence tolerance ε_eq, equivalence sample size K_eq, AVP tolerance ε_AVP. Three-state decomposition: `mut(S) = killed ∪ equiv ∪ survive` (disjoint). SMS formula:

```
SMS_{i,k,j} = |killed_{i,k,j}| / (|mut_j(S_i)| - |equiv_{i,k,j}|)
```

Appendix .7.2. G.2 Degenerate-limit definition (R-8 + P1-3 revision: 3 joint conditions)

The degenerate limit `L = L_equiv ∧ L_killed ∧ L_mut` consists of three **joint conditions** (paired axes), each controlling one layer of the SMS formula. L1-L6 are not 6 independent axes; the pairing responds to dependency queries (R0 W8 / R1 §4 / R2 W3 / DA-MAJOR-3).

Joint condition	Component axes	Pairing rationale
L_equiv (Lemma 9.1)	L1: ε_eq $\rightarrow$ 0; L2: K_eq $\rightarrow \infty$	L1 alone: equiv stays probabilistic (K_eq cannot cover D_S). L2 alone: bitwise equality diluted by ε_eq. Both simultaneously degenerate equiv to classical behavioural equivalence outside a D_S-measure-zero set.
L_killed (Lemma 9.2)	L3: ε_AVP^k $\rightarrow$ 0; L4: MP = {MP_eq}, R(y,y') ≡ y = y'	L3 alone: ε_AVP $\rightarrow$ 0 still allows non-trivial MP (monotonicity, convergence). L4 alone: equality still carries ε_AVP tolerance. Both simultaneously degenerate killed to classical difference detection.

Joint condition	Component axes	Pairing rationale
L_mut (Lemma 9.3)	L5: mut_j $\to$ rule-based syntactic (Mothra AOR/ROR/SDL/CRP); L6: cls(I) $\subseteq$ {imperative deterministic}	L5 alone: syntactic ops on probabilistic/ML may still trigger domain-semantic subsets. L6 alone: imperative programs still mutable by OS/HP/TF/SI. Both simultaneously degenerate mut(S) to Jia & Harman syntactic mutant set.

Appendix .7.3. G.3 Lemmas — Three-state decomposition degenerates under L

Lemma 9.1 (equiv degeneration; P1-3 revision with measure-zero qualification). Under L_equiv (L1 $\wedge$ L2), semantic-class equivalence (E1 $\wedge$ E2) degenerates to classical behavioural equivalence **almost everywhere** w.r.t. measure D_S.

Proof. E1 (type consistency) holds trivially under ε_eq $\to$ 0 (L6 makes imperative output spaces scalar/vector with static types). E2 (|S_i(x) − s'(x)| < ε_eq for K_eq samples) under L1 (ε_eq $\to$ 0) $\wedge$ L2 (K_eq $\to \infty$ with measure-equivalent sampling) is almost-everywhere equivalent to $\forall$ x $\in$ D_S N: S_i(x) = s'(x), where N is a D_S-measure-zero set (continuous D_S: floating-point pathological points / NaN propagation; discrete D_S: N = $\emptyset$). This matches Jia & Harman (2011) §3 classical equivalent-mutant definition under measure-zero equivalence classes. ■

Lemma 9.2 (killed degeneration). Under L3 ∧ L4, killed degenerates to classical difference detection.

Proof. L4 restricts MP to {MP_eq} with R(y, y') ≡ y = y'. For mr = (r, R) and mutant s', the MP_eq violation condition ∃ x: S_i(x) ≠ s'(r(x)) under L3 (ε_AVP → 0) becomes exact inequality. With r = id, violation is S_i(x) ≠ s'(x) — classical difference detection. With r ≠ id, MP_eq still requires S_i(x) = s'(r(x)) as reference oracle from original program; no new state classifications introduced. ■

Lemma 9.3 (mut degeneration). Under L5 ∧ L6, mut_j(S_i) degenerates to the syntactic mutant set of Jia & Harman (2011).

Proof. L5 switches mut_j to rule-based syntactic operators (AOR, ROR, SDL, CRP, UOI — standard Mothra/Proteum sets); L6 restricts PUTs to imperative deterministic programs, excluding triggering conditions for semantic operators on probabilistic/ML programs. Under this configuration, mut_j(S_i) is the literature-defined syntactic mutant set, independent of domain semantics. ■

Appendix .7.4. G.4 Theorem 9.1 — Detailed proof

Theorem 9.1 (SMS → MS degeneration). In the degenerate limit `L = L_equiv ∧ L_killed ∧ L_mut`, almost everywhere w.r.t. D_S,

```
SMS_{i,k,j}  -L→  MS_{i,j} := |killed_{i,j}^classic| / (|mut_j^syntax(
```

Proof. Combine Lemmas 9.1-9.3:

- Numerator: under L3 ∧ L4, killed_{i,k,j} → killed_{i,j}^classic (Lemma 9.2); L4 makes MR_{i,k} trivial over k (only MP_eq remains), so subscript k degenerates.

- Denominator |mut_j(S_i)|: under L5 $\wedge$ L6 $\rightarrow$ |mut_j^syntax(S_i)| (Lemma 9.3).
- Denominator |equiv_{i,k,j}|: under L1 $\wedge$ L2 $\rightarrow$ |equiv_{i,j}^classic| (Lemma 9.1; almost-everywhere w.r.t. D_S).

Substituting into the SMS formula yields MS_{i,j}. ■

Corollary 9.1 (LRCA trivialisation). Under L = L1 $\wedge$ L2 $\wedge$ L3, C = {C1, …, C5} degenerates to {C1}.

Sketch. Each of C2-C5's triggering precondition depends on at least one L_j being violated. When L1 $\wedge$ L2 $\wedge$ L3 hold simultaneously, every non-trivial-space dimension (MP non-triviality, AVP tolerance non-zero, non-empty equiv set, MR-design DOF, class-mapping openness) closes; C2-C5 triggering set becomes empty (read off from §A.2 decision tree). The per-C_k to per-L_j minimum-sufficient mapping depends on §4.6 LRCA-classifier engineering thresholds; we do not claim one-to-one correspondence at formal level — readers can trace via §A.2 + §C.4. Under L, suspect_share $\rightarrow$ 0, LRCA reports only C1 — SMS degenerates to single-layer metric consistent with Jia & Harman (2011) MS engineering structure. ■

Empirical consistency. Theorem 9.1 + Corollary 9.1 jointly guarantee any SMS-based empirical conclusion (Cliff's delta §5.3, Friedman chi^2 §5.5, Spearman rho §5.6) is structurally consistent with existing Jia & Harman (2011) literature in classical syntactic-mutation scenarios — no metric-level semantic fragmentation.